Week-2

Develop an application that receives user‟s name, contact and city and displays the same using Layout Managers and Event Listeners.

Android app in Java using Android Studio that receives the user's **name**, **contact**, and **city** via `EditText` fields and displays them in a `TextView` when a button is clicked.

Using **LinearLayout** as the layout manager and implement an **OnClickListener** for the button.

activity_main.xml

layout with 3 EditTexts, a Button, and a TextView to display the result.

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:padding="16dp"

    android:orientation="vertical">


    <EditText

        android:id="@+id/editTextName"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:hint="Enter Name"

        android:inputType="textPersonName" />


    <EditText

        android:id="@+id/editTextContact"
```

```
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter Contact"
        android:inputType="phone" />

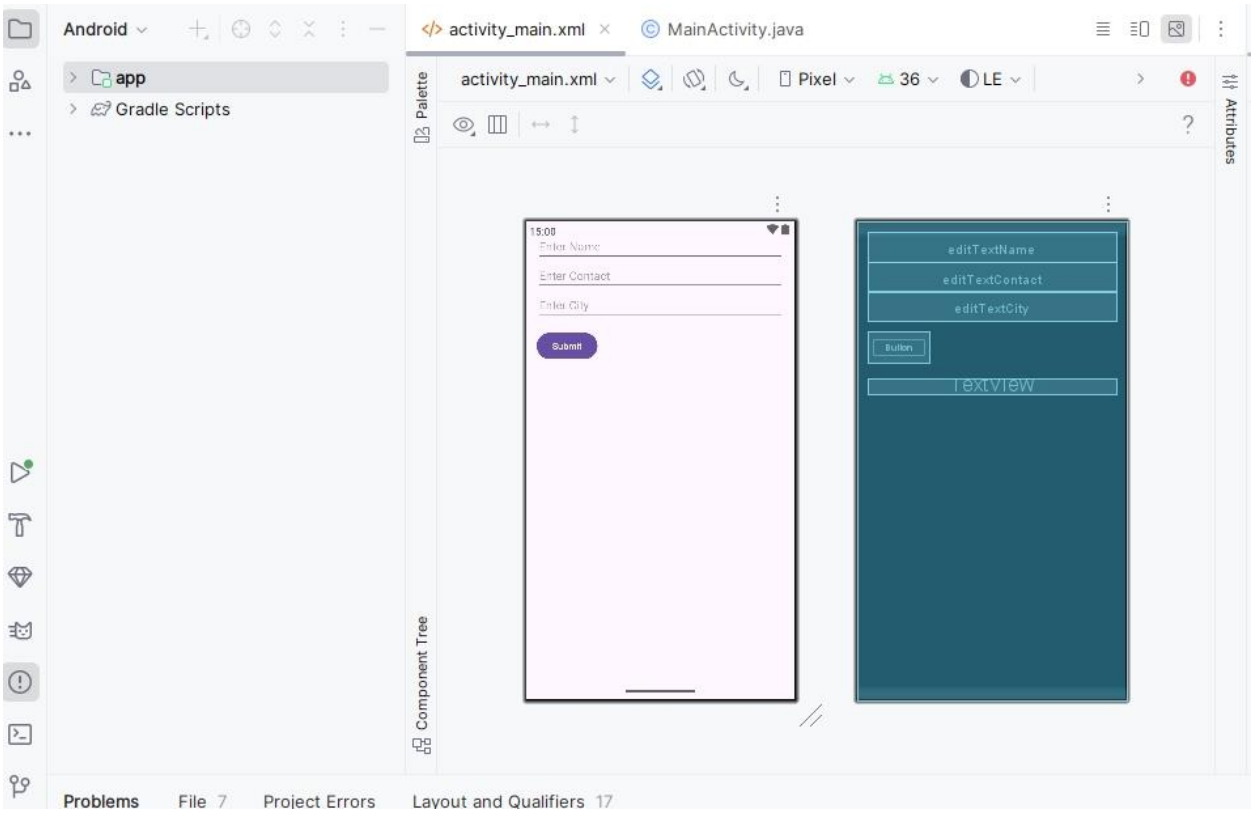
    <EditText
        android:id="@+id/editTextCity"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter City"
        android:inputType="text" />

    <Button
        android:id="@+id/buttonSubmit"
        android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Submit"
        android:layout_marginTop="16dp" />

    <TextView
        android:id="@+id/textViewDisplay"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text=""
```

```
        android:textSize="18sp"
        android:layout_marginTop="24dp" />


</LinearLayout>
```



MainActivity.java

```
package com.example.userinputapp;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle; import
android.view.View; import
android.widget.Button; import
android.widget.EditText; import
android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    EditText editTextName, editTextContact, editTextCity;
    Button buttonSubmit;
    TextView textViewDisplay;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

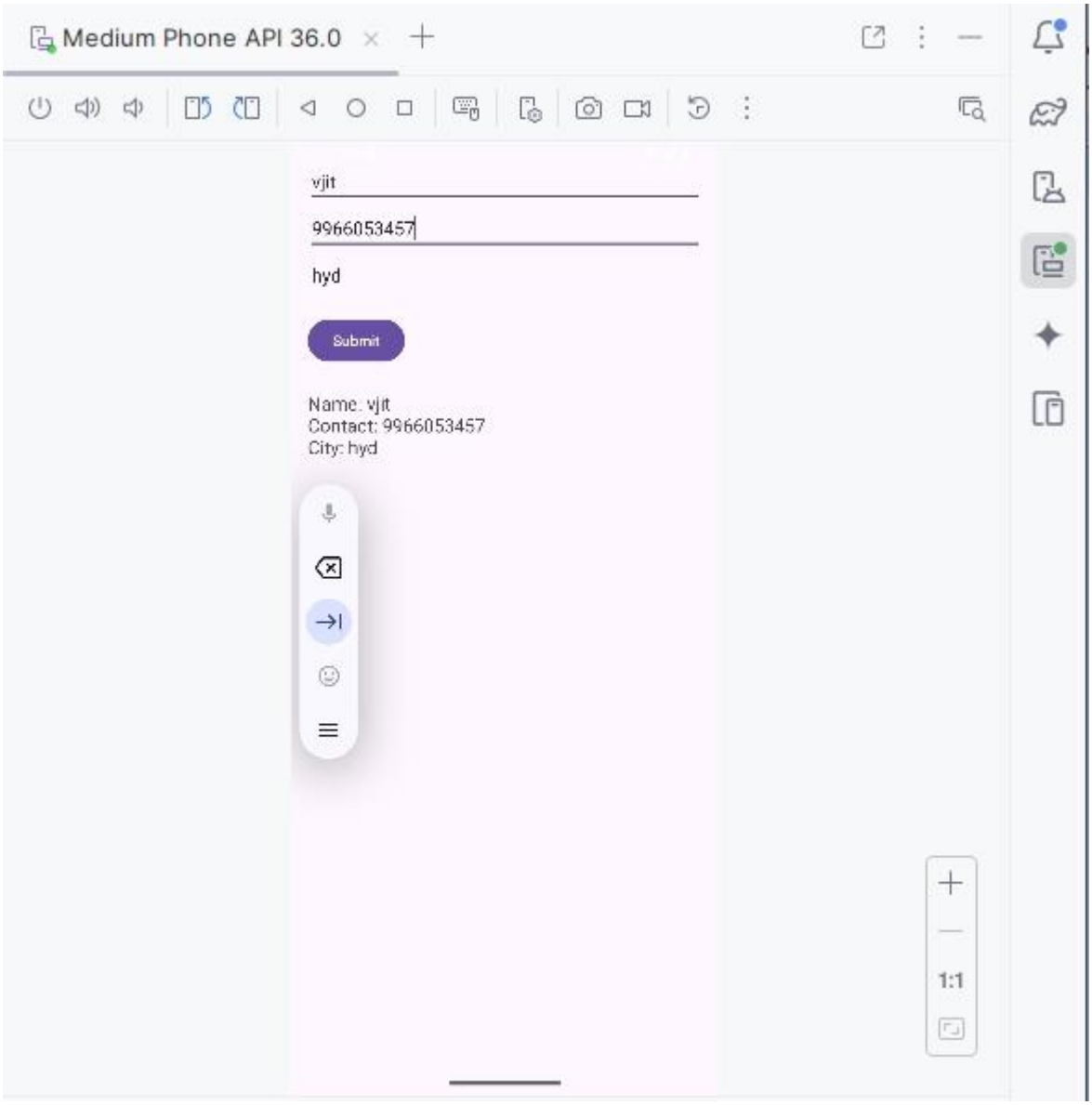
         // Initialize views
        editTextName = findViewById(R.id.editTextName);
        editTextContact = findViewById(R.id.editTextContact);
        editTextCity = findViewById(R.id.editTextCity);
```

```
        buttonSubmit = findViewById(R.id.buttonSubmit);
        textViewDisplay = findViewById(R.id.textViewDisplay);

        // Set listener for button
        buttonSubmit.setOnClickListener(new View.OnClickListener() {
@Override
            public void onClick(View view) {
                // Get input values
                String name = editTextName.getText().toString().trim();
                String contact = editTextContact.getText().toString().trim();
String city = editTextCity.getText().toString().trim();

                // Display the values
                String displayText = "Name: " + name + "\nContact: " + contact + "\nCity: " + city;
textViewDisplay.setText(displayText);
            }
        });
    }
}
```

Output:


Medium Phone API 36.0
vjit
9966053457
hyd
Submit
Name: vjit
Contact: 9966053457
City: hyd
1:1

## Explanation:

- **Layout Manager:** `LinearLayout` arranges the views vertically.
- **Event Listener:** `buttonSubmit.setOnClickListener` listens for the button click.
- When the button is clicked, it fetches the input values and shows them in the `TextView`.

Develop an application that shows names as a list and on selecting a name it should show the details of the candidate on the next screen with a “Back” button.

create a simple **two-screen Android app** in Java using **Android Studio.**

## activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:orientation="vertical"
   android:padding="16dp"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <ListView
      android:id="@+id/listViewNames"
      android:layout_width="match_parent"
      android:layout_height="match_parent" />

</LinearLayout>
```

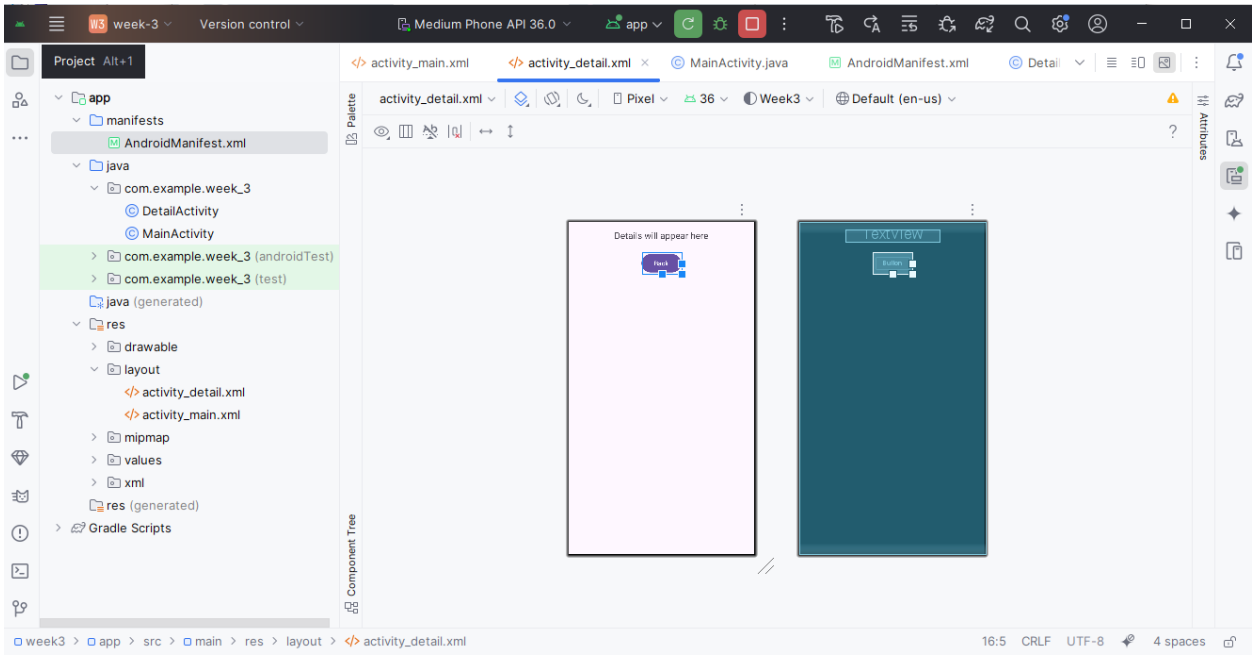
## activity_detail.xml

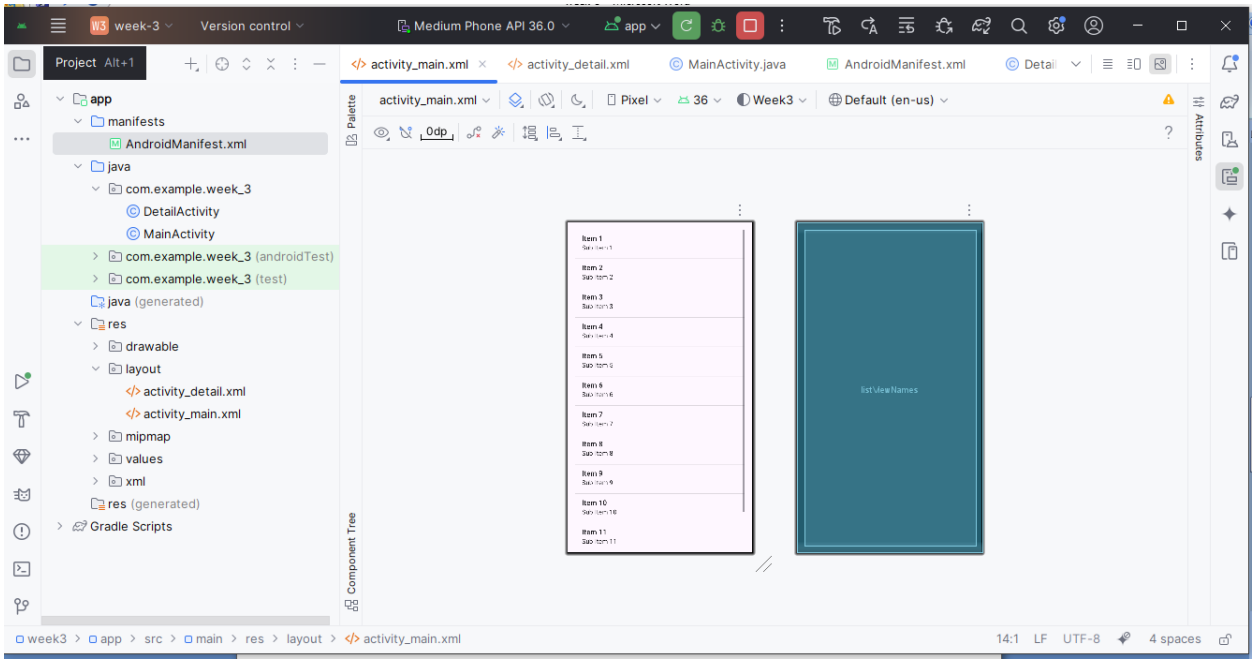
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center_horizontal"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textViewDetails"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:text="Details will appear here" />

    <Button
        android:id="@+id/buttonBack"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Back"
        android:layout_marginTop="24dp" />

</LinearLayout>
```




**MainActivity.java**

```
java
CopyEdit
package com.example.nameapp;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;

public class MainActivity extends AppCompatActivity {

    ListView listViewNames;
    String[] names = {"Alice", "Bob", "Charlie", "David"};
    String[] details = {
        "Alice - Android Developer from NYC",
        "Bob - Data Scientist from LA",
        "Charlie - UX Designer from Chicago",
        "David - QA Engineer from Seattle"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listViewNames = findViewById(R.id.listViewNames);
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_list_item_1, names);
        listViewNames.setAdapter(adapter);

        listViewNames.setOnItemClickListener((parent, view, position, id) ->
{
            Intent intent = new Intent(MainActivity.this,
DetailActivity.class);
            intent.putExtra("details", details[position]);
            startActivity(intent);
        });
    }
}
```

**DetailActivity.java**

```
java
CopyEdit
package com.example.nameapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;

public class DetailActivity extends AppCompatActivity {
```

```
    TextView textViewDetails;
    Button buttonBack;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_detail);

        textViewDetails = findViewById(R.id.textViewDetails);
        buttonBack = findViewById(R.id.buttonBack);

        // Get passed details from Intent
        String detail = getIntent().getStringExtra("details");
        textViewDetails.setText(detail);

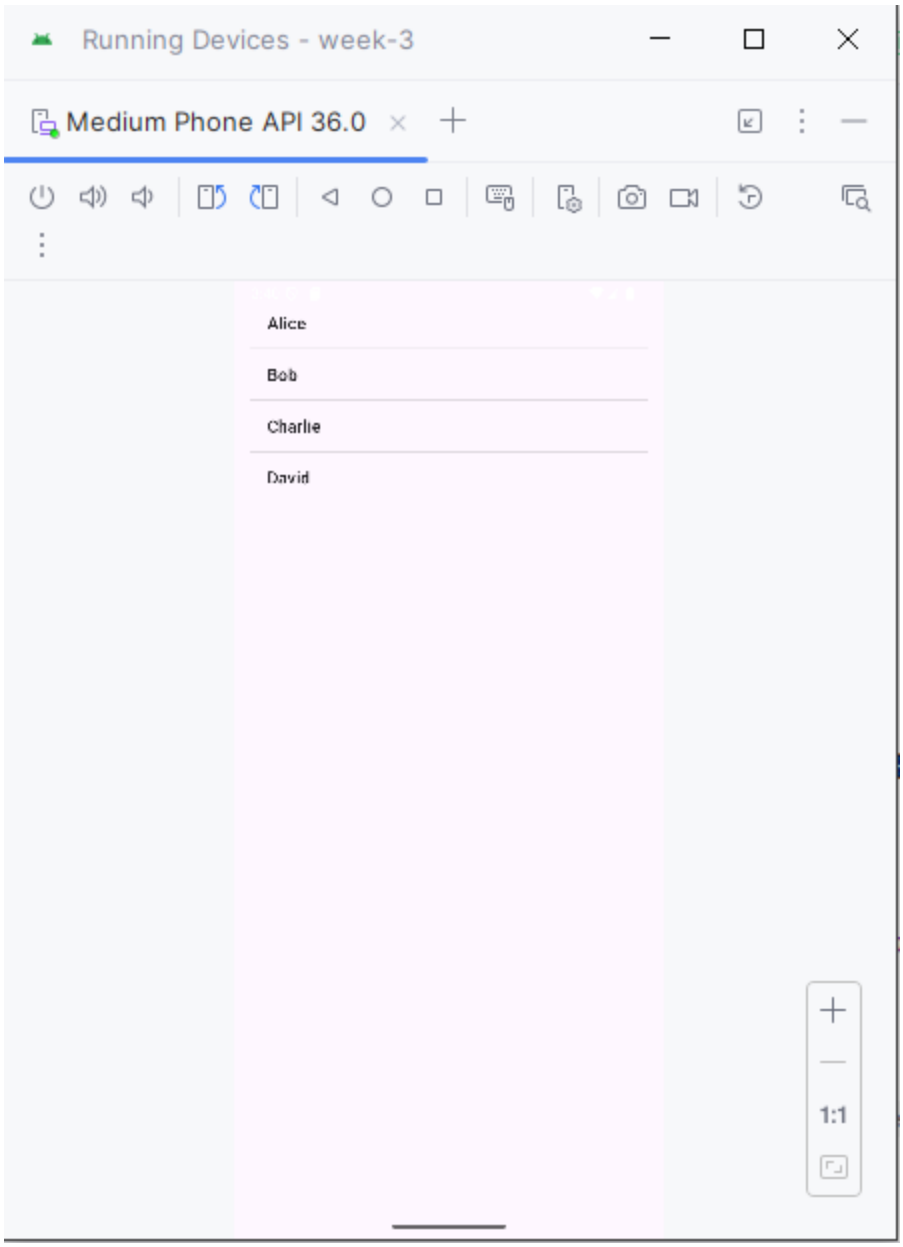
        buttonBack.setOnClickListener(v -> finish()); // Go back to previous
screen
    }
}
```

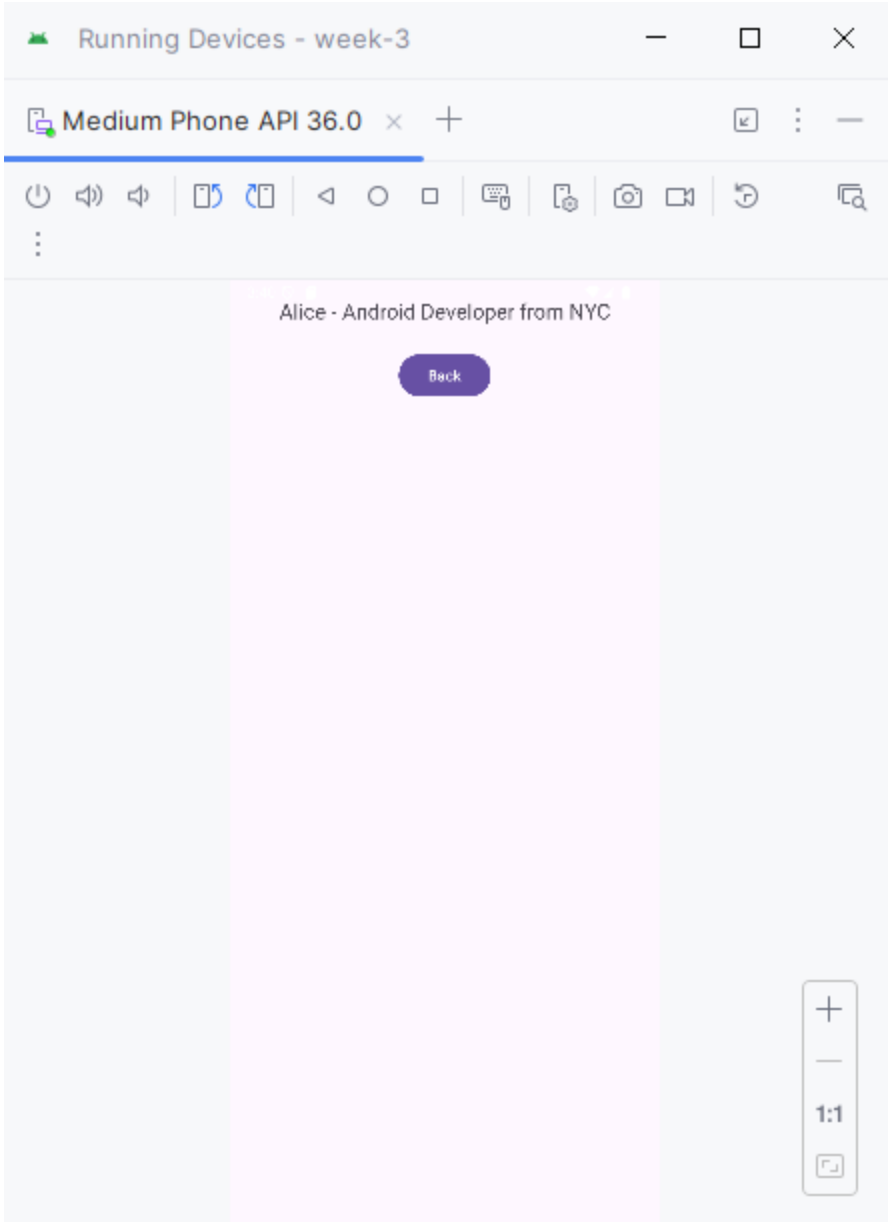
**AndroidManifest.xml**

```
xml
CopyEdit
<application
    ... >

    <activity android:name=".DetailActivity" />
    <activity android:name=".MainActivity">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>

</application>
```


Running Devices - week-3
Medium Phone API 36.0
Alice
Bob
Charlie
David
1:1


Running Devices - week-3
Medium Phone API 36.0
Alice - Android Developer from NYC
Back
1:1

Week-4

Create a native calculator application.

## activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center">

    <EditText
        android:id="@+id/editTextNumber1"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter first number"
        android:inputType="numberDecimal" />

    <EditText
        android:id="@+id/editTextNumber2"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter second number"
        android:inputType="numberDecimal"
        android:layout_marginTop="8dp" />

    <LinearLayout
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="horizontal"
```

```
        android:layout_marginTop="16dp"
        android:gravity="center">

        <Button
            android:id="@+id/btnAdd"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="+" />

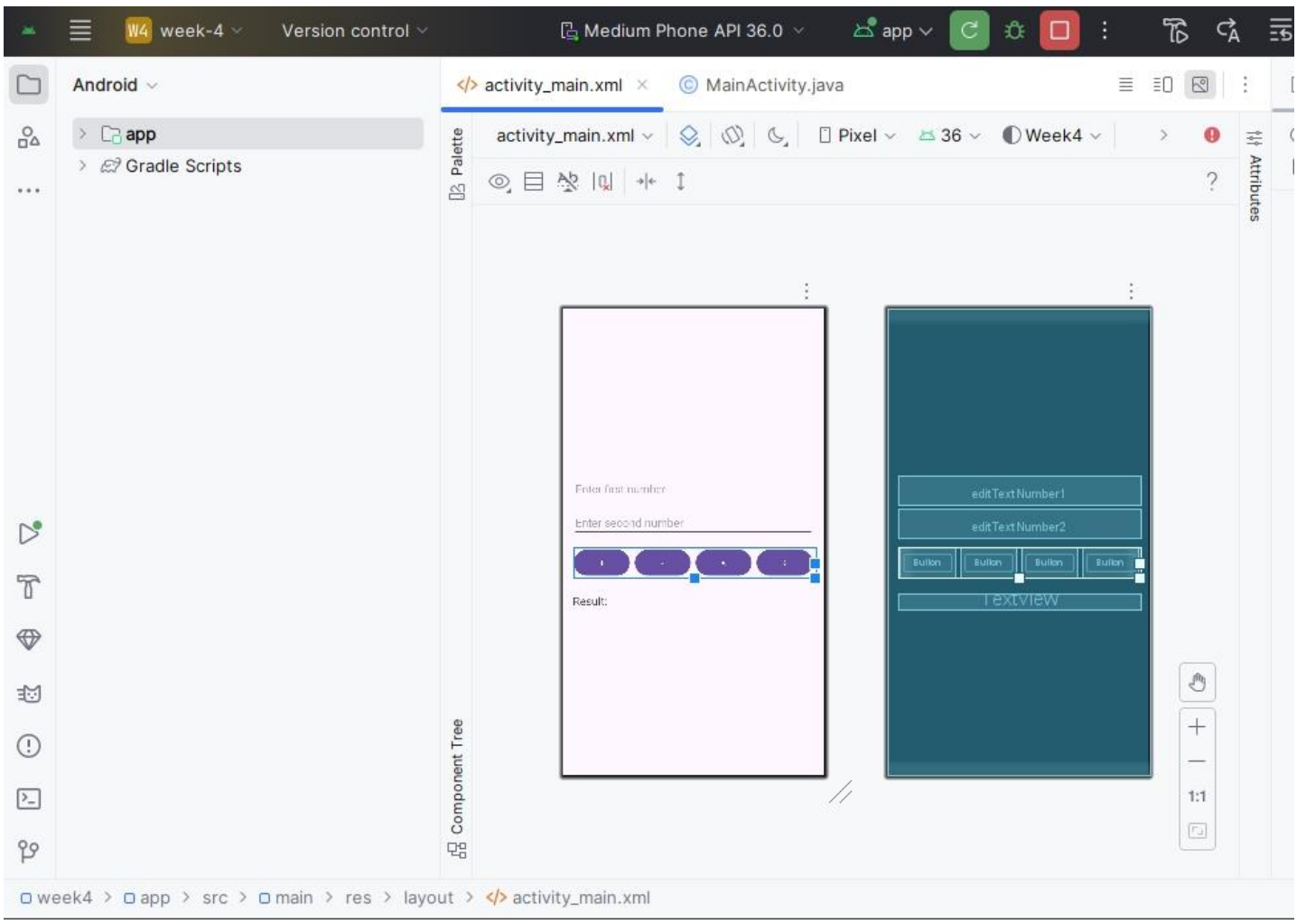
        <Button
            android:id="@+id/btnSubtract"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="-"
            android:layout_marginStart="8dp" />

        <Button
            android:id="@+id/btnMultiply"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="×"
            android:layout_marginStart="8dp" />

        <Button
            android:id="@+id/btnDivide"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="÷"
            android:layout_marginStart="8dp" />
    </LinearLayout>
```

```
   <TextView
       android:id="@+id/textViewResult"
       android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Result: "
       android:textSize="18sp"
       android:layout_marginTop="24dp" />

</LinearLayout>
```




# MainActivity.java

```
package com.example.calculatorapp;


import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

```
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    EditText editTextNumber1, editTextNumber2;
    Button btnAdd, btnSubtract, btnMultiply, btnDivide;
    TextView textViewResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editTextNumber1 = findViewById(R.id.editTextNumber1);
        editTextNumber2 = findViewById(R.id.editTextNumber2);
        btnAdd = findViewById(R.id.btnAdd);
        btnSubtract = findViewById(R.id.btnSubtract);
        btnMultiply = findViewById(R.id.btnMultiply);
        btnDivide = findViewById(R.id.btnDivide);
        textViewResult = findViewById(R.id.textViewResult);

        btnAdd.setOnClickListener(v -> calculate('+'));
        btnSubtract.setOnClickListener(v -> calculate('-'));
btnMultiply.setOnClickListener(v -> calculate('*'));
btnDivide.setOnClickListener(v -> calculate('/'));
    }

    private void calculate(char operator) {
```

```
        String num1Str = editTextNumber1.getText().toString();
        String num2Str = editTextNumber2.getText().toString();

        if (num1Str.isEmpty() || num2Str.isEmpty()) {
            textViewResult.setText("Please enter both numbers");
return;
        }

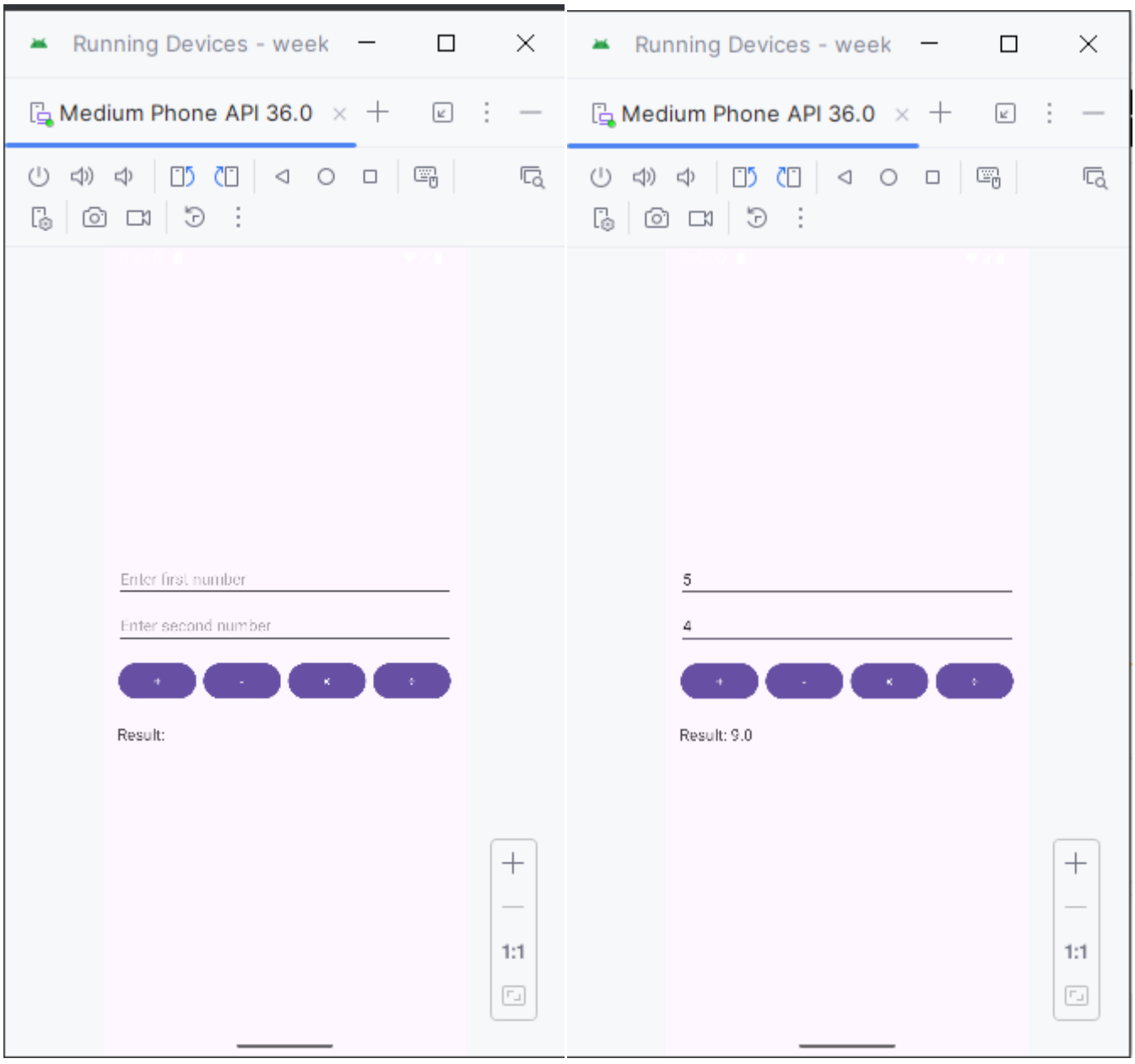
        double num1 = Double.parseDouble(num1Str);
        double num2 = Double.parseDouble(num2Str);
        double result = 0;

        switch (operator) {
            case '+':
                result = num1 + num2;
                break;
            case '-':
                result = num1 - num2;
                break;
            case '*':
                result = num1 * num2;
                break;
            case '/':
                if (num2 == 0) {
                    textViewResult.setText("Cannot divide by zero");
return;
                }
                result = num1 / num2;
                break;
```

```
        }

        textViewResult.setText("Result: " + result);
    }
}
```

**Output:**



Week-5

Create an application that saves user information like name, age, gender etc. in shared preference and retrieves them when the program restarts.

## Functionality

1. User enters `name`, `age`, and `gender`.
2. Info is saved using **SharedPreferences**.
3. When the app restarts, data is automatically loaded from preferences.

## activity_main.xml (UI Layout)

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">


   <EditText

      android:id="@+id/editName"

      android:layout_width="match_parent"

android:layout_height="wrap_content"

android:hint="Name" />


   <EditText

      android:id="@+id/editAge"

      android:layout_width="match_parent"

android:layout_height="wrap_content"

android:inputType="number"

      android:hint="Age" />


   <EditText

      android:id="@+id/editGender"

      android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:hint="Gender" />

    <Button
        android:id="@+id/btnSave"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Save Info" />

    <TextView
        android:id="@+id/textDisplay"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingTop="20dp"
        android:text="User Info Will Be Displayed Here" />
</LinearLayout>
```

# MainActivity.java

```
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText editName, editAge, editGender;
Button btnSave;
    TextView textDisplay;
```

```
    SharedPreferences sharedPreferences;
    public static final String PREF_NAME = "UserInfo";


    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        editName = findViewById(R.id.editName);
        editAge = findViewById(R.id.editAge);
        editGender = findViewById(R.id.editGender);
btnSave = findViewById(R.id.btnSave);
        textDisplay = findViewById(R.id.textDisplay);


        sharedPreferences = getSharedPreferences(PREF_NAME, MODE_PRIVATE);


loadUserData();


        btnSave.setOnClickListener(v -> {
saveUserData();
        });
    }


    private void saveUserData() {
        String name = editName.getText().toString();
        String age = editAge.getText().toString();
        String gender = editGender.getText().toString();


        SharedPreferences.Editor editor = sharedPreferences.edit();
editor.putString("Name", name);
        editor.putString("Age", age);
```

```
        editor.putString("Gender", gender);
        editor.apply();


        textDisplay.setText("Saved: " + name + ", " + age + ", " + gender);
    }


    private void loadUserData() {
        String name = sharedPreferences.getString("Name", "Not set");
        String age = sharedPreferences.getString("Age", "Not set");
        String gender = sharedPreferences.getString("Gender", "Not set");


        textDisplay.setText("Loaded: " + name + ", " + age + ", " + gender);
    }
}
```


Running Devices - week
Medium Phone API 36.0
Name
Age
Gender
Save Info
Saved: uday123, 40, Male
1:1

Week-6

## Develop a Registration and Login application that makes use of database.

The SQlite database as its name suggests, is a lightweight database which is used to store the data locally in your Android mobile devices. One can create a simple login and register app using SQlite which can also work as test bench or platform for launching the app for web servers.

Let us first create a Database helper class which extends the properties of SQliteOpenHelperClass.

The code snippet for DB Helper class "DBHelper.class" is as follows:

```
package com.learnandroid.loginsqlite;
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import androidx.annotation.Nullable;

public class DBHelper  extends SQLiteOpenHelper {
public static final String DBNAME = "Login.db";
public DBHelper(Context context) {
        super(context, "Login.db", null, 1);
}

    @Override
    public void onCreate(SQLiteDatabase MyDB) {
        MyDB.execSQL("create Table users(username TEXT primary key, password
TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase MyDB, int i, int i1) {
MyDB.execSQL("drop Table if exists users");
    }

    public Boolean insertData(String username, String password){
SQLiteDatabase MyDB = this.getWritableDatabase();
        ContentValues contentValues= new ContentValues();
        contentValues.put("username", username);
        contentValues.put("password", password);          long
result = MyDB.insert("users", null, contentValues);
```

```
if(result==-1) return false;
          else
        return true;
    }

    public Boolean checkusername(String username) {
        SQLiteDatabase MyDB = this.getWritableDatabase();
        Cursor cursor = MyDB.rawQuery("Select * from users where username =
?", new String[]{username});
        if (cursor.getCount() > 0)
            return true;
        else
        return false;
    }

    public Boolean checkusernamepassword(String username, String
password){
        SQLiteDatabase MyDB = this.getWritableDatabase();
        Cursor cursor = MyDB.rawQuery("Select * from users where username = ?
and password = ?", new String[] {username,password});
        if(cursor.getCount()>0)
            return true;
        else
        return false;
    }
}
```

Let us first create main_activity.xml which serves as the register page. This page will have three edit text boxes, username, password and retype password. The code snippet for main_activity.xml is as follows.

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:padding="10dp"
  tools:context=".MainActivity">

    <EditText
    android:id="@+id/username"
    android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="User Name"
    android:layout_marginTop="50dp"/>
```

```
    <EditText
    android:id="@+id/password"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:layout_marginTop="50dp"
    android:layout_below="@+id/username"/>

    <EditText
    android:id="@+id/repassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Retype Password"
    android:layout_marginTop="50dp"
    android:layout_below="@+id/password"/>

    <Button
    android:id="@+id/btnsignup"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Register"
    android:layout_marginTop="50dp"
    android:layout_below="@+id/repassword"/>

    <Button
    android:id="@+id/btnsignin"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Existing user! Go to Sign in page"
android:layout_marginTop="50dp"
    android:layout_below="@+id/btnsignup"/>

</RelativeLayout>
```

The code snippet for MainActivity.java is as follows.

```
package com.learnandroid.loginsqlite;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    EditText username, password, repassword;
```

```
    Button signup, signin;
    DBHelper DB;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        username = (EditText) findViewById(R.id.username);
        password = (EditText) findViewById(R.id.password);
        repassword = (EditText) findViewById(R.id.repassword);
signup = (Button) findViewById(R.id.btnsignup);
        signin = (Button) findViewById(R.id.btnsignin);
        DB = new DBHelper(this);

        signup.setOnClickListener(new View.OnClickListener() {
@Override
            public void onClick(View view) {
                String user = username.getText().toString();
                String pass = password.getText().toString();
                String repass = repassword.getText().toString();

                if(user.equals("")||pass.equals("")||repass.equals(""))
Toast.makeText(MainActivity.this, "Please enter all the fields",
Toast.LENGTH_SHORT).show();
                else{
                    if(pass.equals(repass)){
                        Boolean checkuser = DB.checkusername(user);
                        if(checkuser==false){
                            Boolean insert = DB.insertData(user,
pass);
                            if(insert==true){
                                Toast.makeText(MainActivity.this,
"Registered successfully", Toast.LENGTH_SHORT).show();
                                Intent intent = new
Intent(getApplicationContext(),HomeActivity.class);
                                startActivity(intent);
                            }else{
                                Toast.makeText(MainActivity.this,
"Registration failed", Toast.LENGTH_SHORT).show();

                            }
                        }
                        else{
                            Toast.makeText(MainActivity.this, "User
already exists! please sign in", Toast.LENGTH_SHORT).show();
                        }
                    }else{
```

```
                        Toast.makeText(MainActivity.this, "Passwords not
matching", Toast.LENGTH_SHORT).show();
                    }
                } }
        });

        signin.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(getApplicationContext(),
LoginActivity.class);
                startActivity(intent);
            }
        });
    }
}
```

Now create another activity for Login. The code snippet for activity_login.xml is as follows.

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:padding="10dp"
  tools:context=".LoginActivity">

    <EditText
    android:id="@+id/username1"
    android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="User Name"
    android:layout_marginTop="50dp"/>

    <EditText
    android:id="@+id/password1"
    android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Password"
    android:layout_marginTop="50dp"

    android:layout_below="@+id/username1"/>

    <Button
    android:id="@+id/btnsignin1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Sign in"
    android:layout_marginTop="50dp"
```

```
    android:layout_below="@+id/password1"/>
</RelativeLayout>
```

The code snippet for LoginActivity.java is as follows:

```
package com.learnandroid.loginsqlite;
import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.os.Bundle; import
android.view.View; import
android.widget.Button; import
android.widget.EditText; import
android.widget.Toast;

public class LoginActivity extends AppCompatActivity {
    EditText username, password;
    Button btnlogin;
    DBHelper DB;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        username = (EditText) findViewById(R.id.username1);
password = (EditText) findViewById(R.id.password1);
btnlogin = (Button) findViewById(R.id.btnsignin1);
DB = new DBHelper(this);

        btnlogin.setOnClickListener(new View.OnClickListener() {
@Override
            public void onClick(View view) {

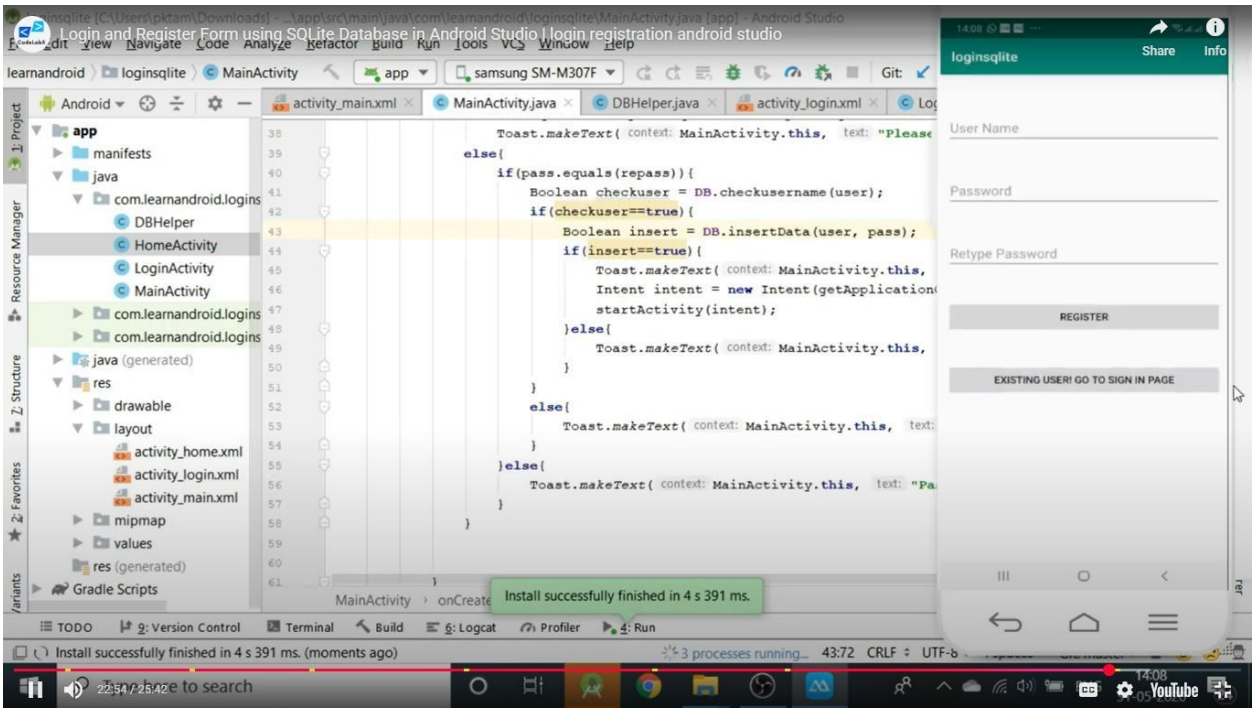
                String user = username.getText().toString();
String pass = password.getText().toString();

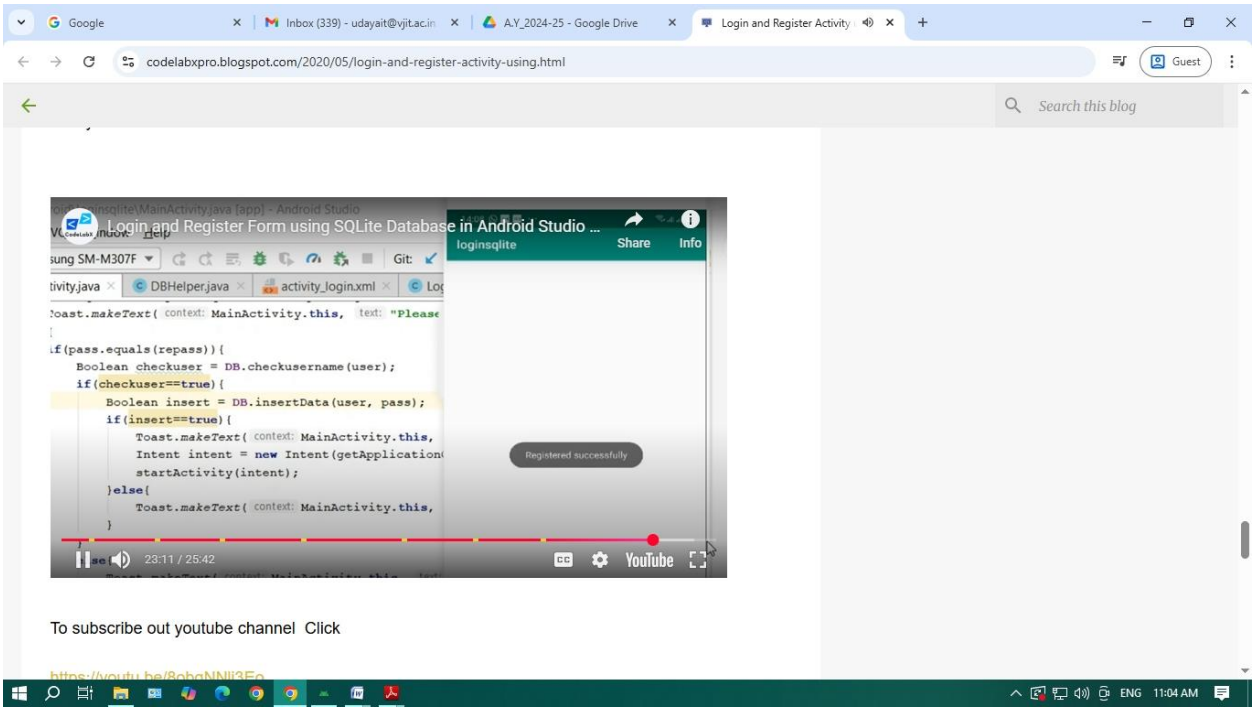
                if(user.equals("")||pass.equals(""))

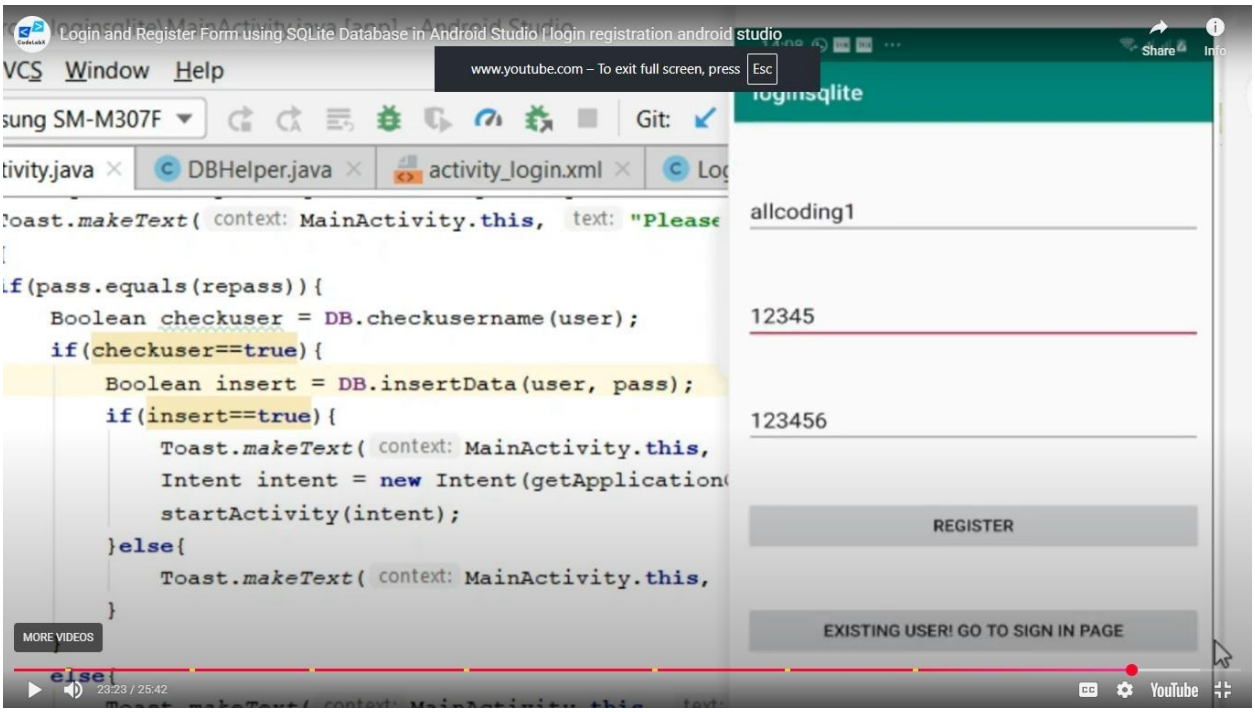
                    Toast.makeText(LoginActivity.this, "Please enter all the
fields", Toast.LENGTH_SHORT).show();
                else{
                    Boolean checkuserpass =
DB.checkusernamepassword(user, pass);
                    if(checkuserpass==true){
                        Toast.makeText(LoginActivity.this, "Sign in
successfull", Toast.LENGTH_SHORT).show();
                        Intent intent = new
Intent(getApplicationContext(), HomeActivity.class);
```

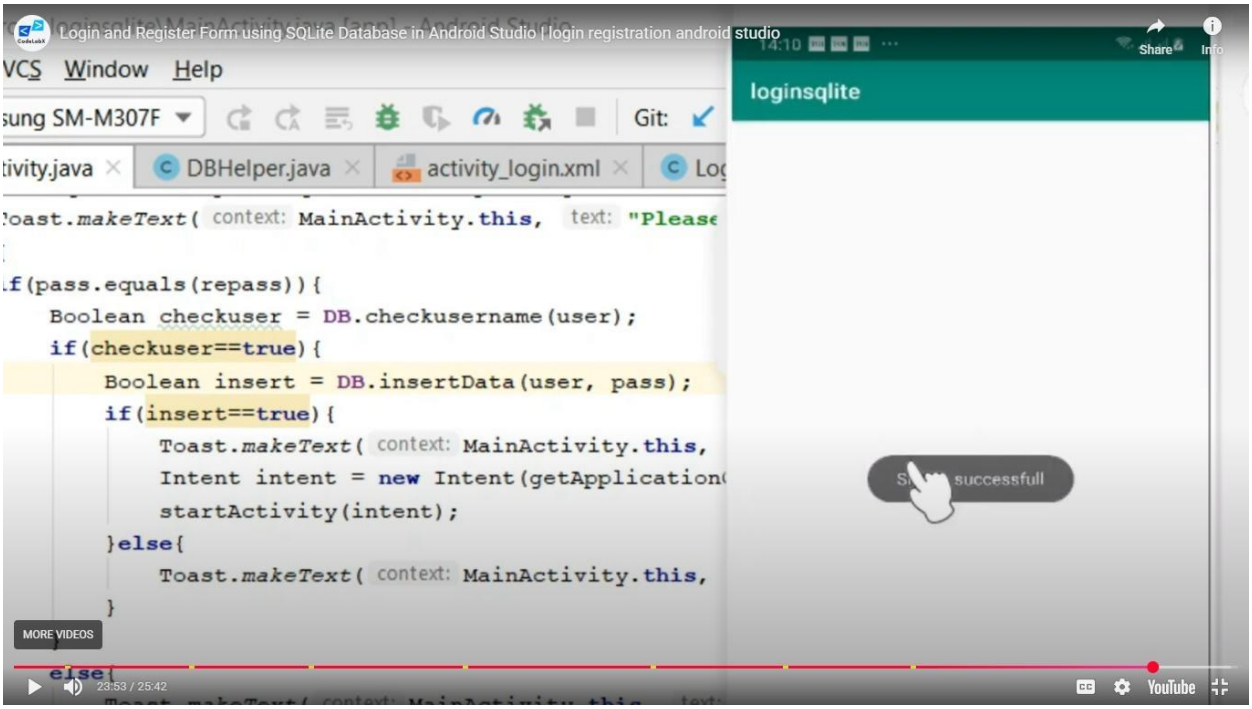
```
                    startActivity(intent);
                }else{
                    Toast.makeText(LoginActivity.this, "Invalid
Credentials", Toast.LENGTH_SHORT).show();
                }
            }
        }
    });
    }
}
```

Output:




codelabxpro.blogspot.com/2020/05/login-and-register-activity-using.html
Search this blog
Login and Register Form using SQLite Database in Android Studio ...
loginsqlite
Share
Info
Registered successfully
23:11 / 25:42
YouTube
To subscribe out youtube channel Click


Login and Register Form using SQLite Database in Android Studio | login registration android studio
www.youtube.com – To exit full screen, press Esc
Share
Info
loginsqlite
allcoding1
12345
123456
REGISTER
EXISTING USER! GO TO SIGN IN PAGE
MORE VIDEOS
23:23 / 25:42
YouTube




Supporting link is “https://codelabxpro.blogspot.com/2020/05/login-and-register-activity-using.html”

**Week 7:**

Develop an application that inserts some notifications into Notification area and whenever a notification is inserted, it should show a toast with details of the notification.

Activtity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">


    <Button
        android:id="@+id/btnCreateNotification"
android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Create Notification"
        tools:layout_editor_absoluteX="170dp"
        tools:layout_editor_absoluteY="322dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Mainactivity.java

```
package com.example.notify;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;

import android.annotation.SuppressLint;
import android.os.Bundle;

/*public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}*/
import android.app.NotificationChannel ;
import android.app.NotificationManager ;
import android.content.ContentResolver ;
import android.content.Context ;
import android.graphics.Color ;
import android.media.AudioAttributes ;
import android.net.Uri ;
//import android.support.v4.app.NotificationCompat ;
//import android.support.v7.app.AppCompatActivity ;
//import android.os.Bundle ;
import android.view.View ;
```

```
import android.widget.Button ;

import java.util.Objects;

public class MainActivity extends AppCompatActivity {
    public static final String NOTIFICATION_CHANNEL_ID = "10001" ;
private final static String default_notification_channel_id;

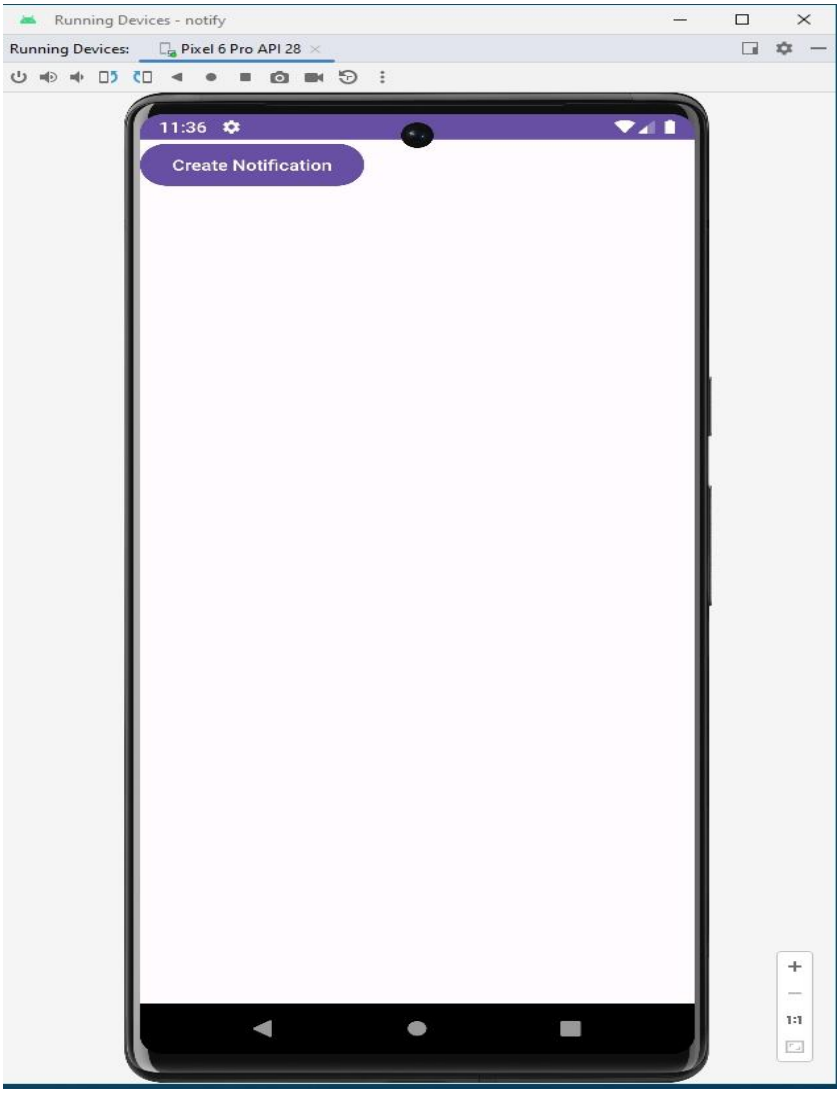
    static {
        default_notification_channel_id = "default";
}

    @Override
    protected void onCreate (Bundle savedInstanceState) {
        super .onCreate(savedInstanceState) ;
        setContentView(R.layout. activity_main ) ;
        @SuppressLint({"MissingInflatedId", "LocalSuppress"}) Button
btnCreateNotification = findViewById(R.id.btnCreateNotification ) ;
        btnCreateNotification.setOnClickListener( new View.OnClickListener()
{
            @SuppressLint("NotificationPermission")
            @Override
            public void onClick (View v) {
                Uri sound = Uri. parse (ContentResolver.
SCHEME_ANDROID_RESOURCE + "://" + getPackageName() +
"/raw/quite_impressed.mp3" ) ;
                NotificationCompat.Builder mBuilder = new
NotificationCompat.Builder(MainActivity. this,
                        default_notification_channel_id )
                        .setSmallIcon(R.drawable. ic_launcher_foreground )
                        .setContentTitle( "Test" )
                        .setSound(sound)
                        .setContentText( "Hello! This is my first push
notification" );
                NotificationManager mNotificationManager =
(NotificationManager) getSystemService(Context. NOTIFICATION_SERVICE ) ;
                if (android.os.Build.VERSION. SDK_INT >=
android.os.Build.VERSION_CODES. O ) {
                    AudioAttributes audioAttributes = new
AudioAttributes.Builder()
                            .setContentType(AudioAttributes.
CONTENT_TYPE_SONIFICATION )
                            .setUsage(AudioAttributes. USAGE_ALARM )
                            .build() ;
                    int importance = NotificationManager. IMPORTANCE_HIGH ;
NotificationChannel notificationChannel = new
                           NotificationChannel( NOTIFICATION_CHANNEL_ID ,
"NOTIFICATION_CHANNEL_NAME" , importance) ;
                    notificationChannel.enableLights( true ) ;
                    notificationChannel.setLightColor(Color. RED ) ;
                    notificationChannel.enableVibration( true ) ;
                    notificationChannel.setVibrationPattern( new long []{ 100
, 200 , 300 , 400 , 500 , 400 , 300 , 200 , 400 }) ;
                    notificationChannel.setSound(sound , audioAttributes) ;
mBuilder.setChannelId( NOTIFICATION_CHANNEL_ID ) ;
                    assert mNotificationManager != null;


mNotificationManager.createNotificationChannel(notificationChannel) ;
```

```
                }
                Objects.requireNonNull(mNotificationManager).notify(( int )
System. currentTimeMillis (), mBuilder.build()) ;
            }
        }) ;
    }
}
```

output:






Week-8

Develop an application that uses a menu with 3 options for dialing a number, opening a website and to send an SMS. On selecting an option, the appropriate action should be invoked using intents.

**Step-by-Step Implementation**

**1. Create a New Android Project**

- Open **Android Studio**
- Select **Empty View Activity**
- Name it (e.g., IntentMenuApp)
- Choose **Language: Java**
- Minimum SDK: API 21 or above.

---

**activity_main.xml**

This layout includes three buttons for the three actions.

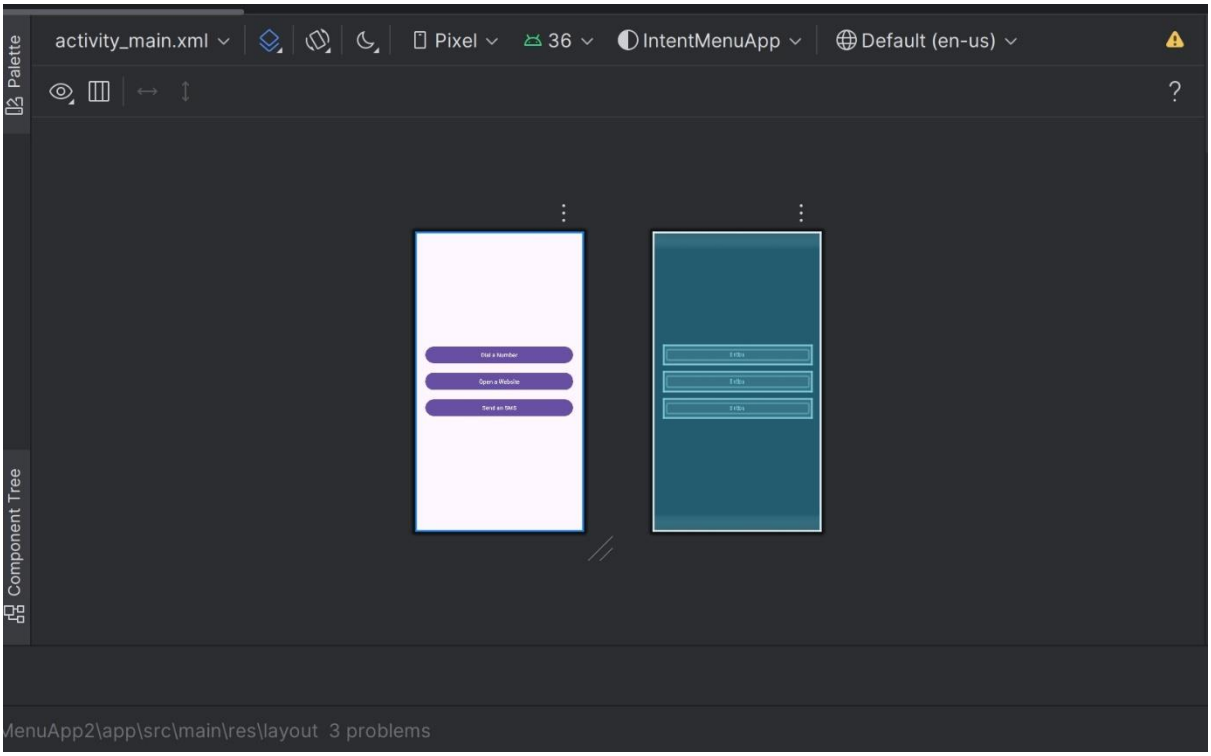
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="24dp">

   <Button
      android:id="@+id/btnDial"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Dial a Number" />
```

```
    <Button
        android:id="@+id/btnWebsite"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Open a Website"
        android:layout_marginTop="16dp"/>

    <Button
        android:id="@+id/btnSMS"
        android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Send an SMS"
        android:layout_marginTop="16dp"/>

</LinearLayout>
```




---

**MainActivity.java**

This Java file contains the logic to handle button clicks and launch the corresponding **Intent**.

```
package com.example.intentmenuapp;


import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;

import android.net.Uri;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.Toast;


public class MainActivity extends AppCompatActivity {


Button btnDial, btnWebsite, btnSMS;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

       btnDial = findViewById(R.id.btnDial);
       btnWebsite = findViewById(R.id.btnWebsite);
       btnSMS = findViewById(R.id.btnSMS);

        // Option 1: Dial a Number
        btnDial.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(Intent.ACTION_DIAL);
                intent.setData(Uri.parse("tel:1234567890")); // sample number
startActivity(intent);
            }
        });

        // Option 2: Open a Website
        btnWebsite.setOnClickListener(new View.OnClickListener() {
@Override
            public void onClick(View v) {
                Intent intent = new Intent(Intent.ACTION_VIEW);
                intent.setData(Uri.parse("https://www.google.com"));
                startActivity(intent);
            }
       });

        // Option 3: Send an SMS
```

```
        btnSMS.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(Intent.ACTION_VIEW);
                intent.setData(Uri.parse("sms:1234567890"));
                intent.putExtra("sms_body", "Hello! This is a test
message.");                try {
                    startActivity(intent);
                } catch (Exception e) {
                    Toast.makeText(MainActivity.this, "No SMS app
found!", Toast.LENGTH_SHORT).show();
                }
}
        });
    }
}
```

---

**AndroidManifest.xml**

You **don't need special permissions** for ACTION_DIAL or ACTION_VIEW with sms: or https: — but if you use ACTION_CALL or send SMS directly, you would.

For this app, you can just leave the default manifest like this:

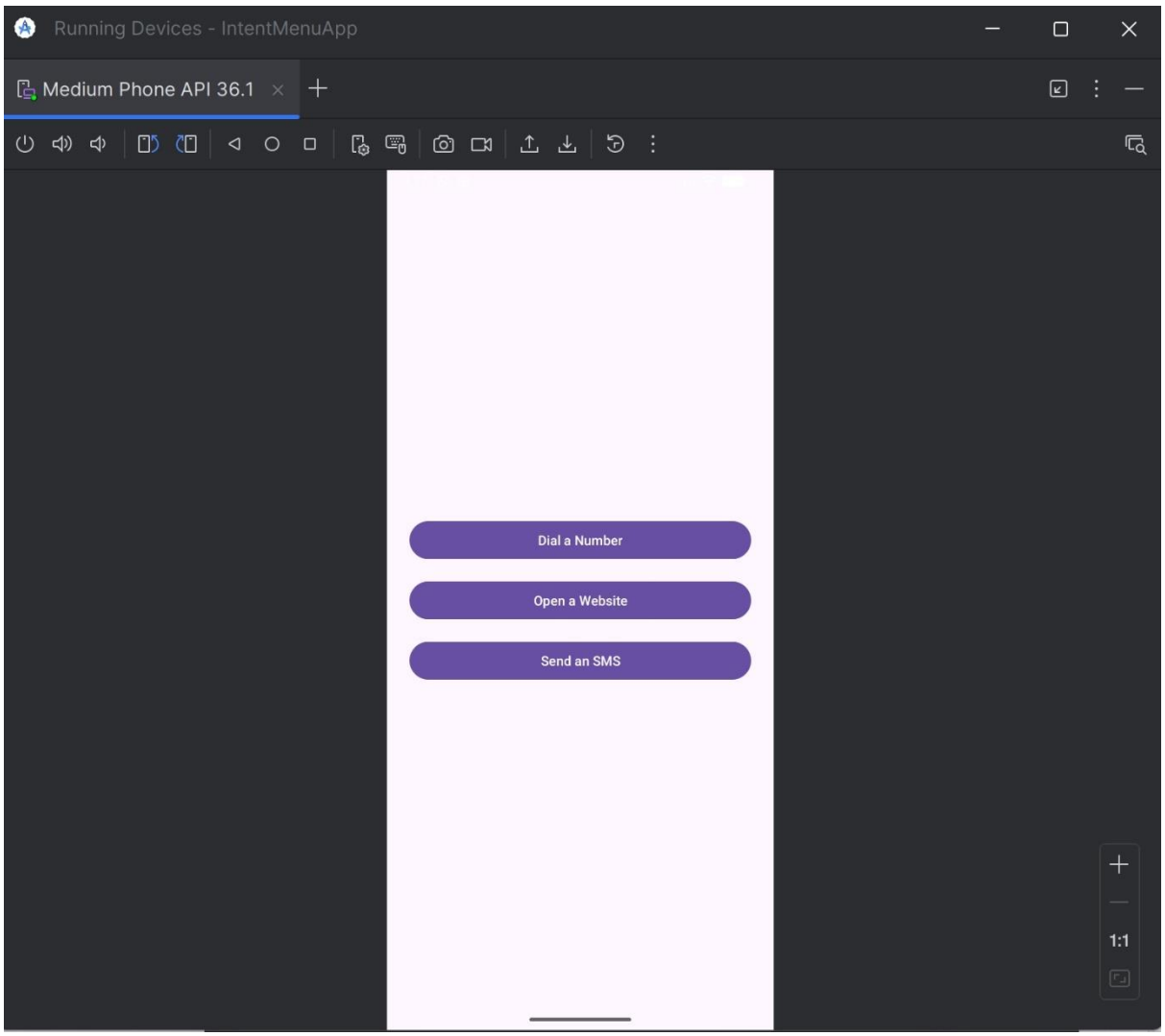
```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
package="com.example.intentmenuapp">

    <application
        android:allowBackup="true"
        android:label="Intent Menu App"
        android:icon="@mipmap/ic_launcher"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
```

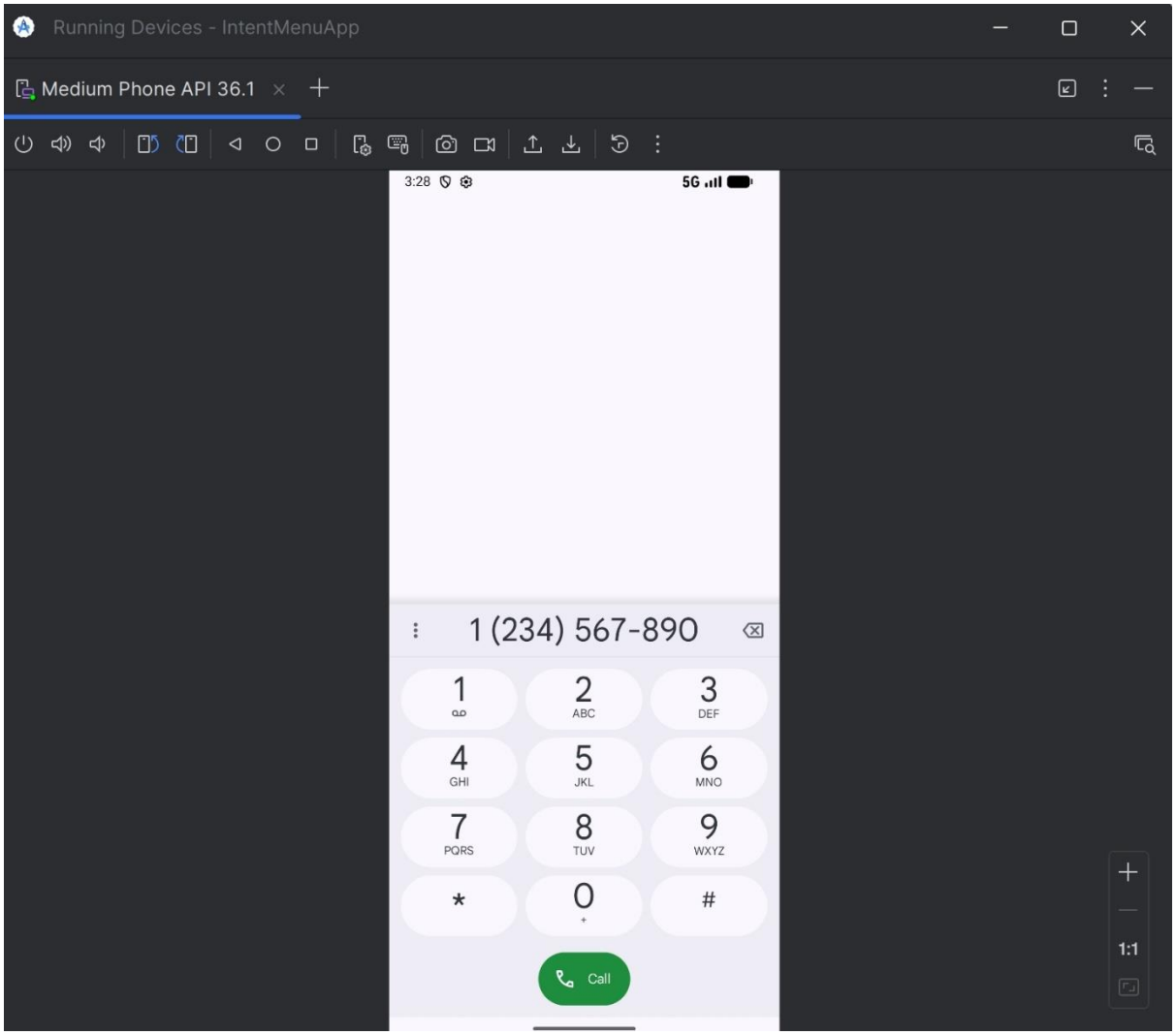
```
        android:theme="@style/Theme.IntentMenuApp">
        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category
android:name="android.intent.category.LAUNCHER"/>          </intent -
filter>
        </activity>
    </application>

</manifest>
```

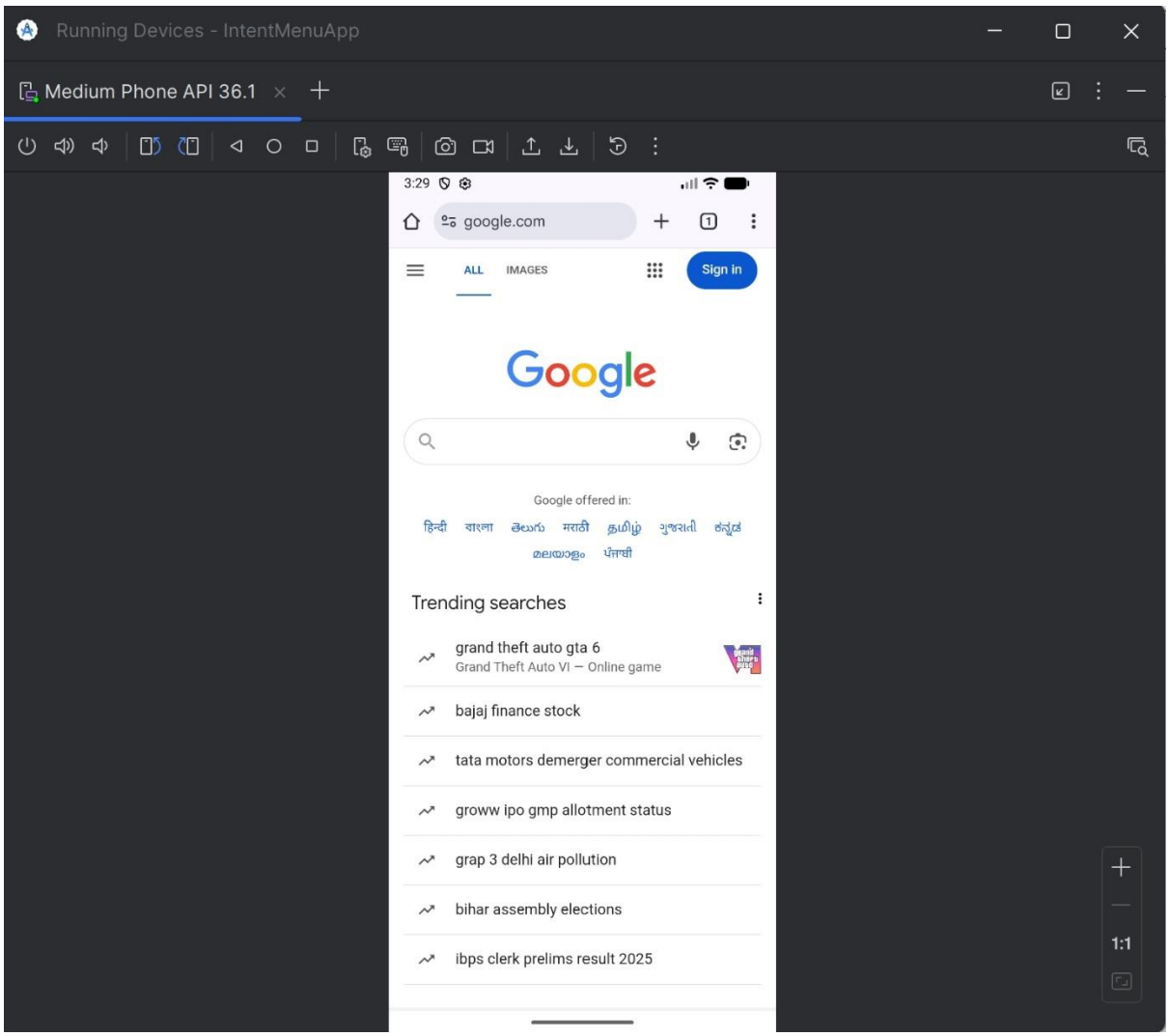
Ouput:


Running Devices - IntentMenuApp
Medium Phone API 36.1
Dial a Number
Open a Website
Send an SMS
1:1

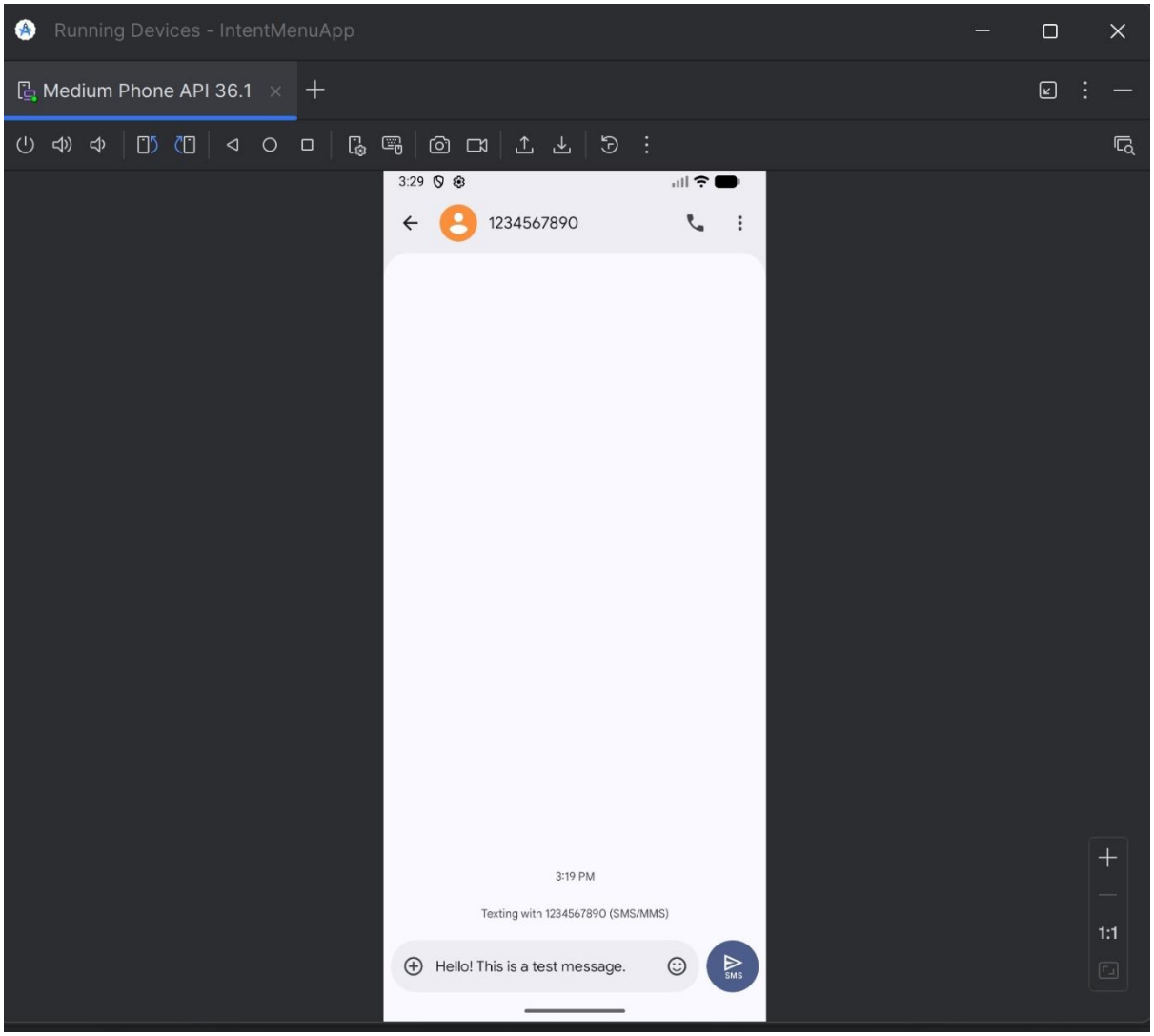
Dial Number:




Open Website:


Running Devices - IntentMenuApp
Medium Phone API 36.1
3:29
google.com
ALL
IMAGES
Sign in
Google
Google offered in:
हिन्दी বাংলা తెలుగు मराठी தமிழ் ગુજરાતી ಕನ್ನಡ
മലയാളം ਪੰਜਾਬੀ
Trending searches
grand theft auto gta 6
Grand Theft Auto VI — Online game
bajaj finance stock
tata motors demerger commercial vehicles
groww ipo gmp allotment status
grap 3 delhi air pollution
bihar assembly elections
ibps clerk prelims result 2025
1:1

SMS:


Running Devices - IntentMenuApp
Medium Phone API 36.1
3:29
1234567890
3:19 PM
Texting with 1234567890 (SMS/MMS)
Hello! This is a test message.
SMS
1:1

Week-9

Create an alarm application using time picker to set alarm time.

**Create the Layout (activity_main.xml)**

```
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp">

   <Button
      android:id="@+id/btnSetTime"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Set Alarm Time" />

   <TextView
      android:id="@+id/tvAlarmTime"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="No Alarm Set" />

</LinearLayout>
```

**MainActivity.java**

```
package com.example.alarmapp;

//package com.example.alarmapp;

import android.app.AlarmManager;
import android.app.PendingIntent;
import android.app.TimePickerDialog;
import android.content.Context; import
android.content.Intent;
import android.os.Bundle;

import android.widget.Button;
```

```
import android.widget.TextView;
import android.widget.TimePicker;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
private TextView tvAlarmTime;
   private AlarmManager alarmManager;
   private PendingIntent pendingIntent;
   private Calendar calendar = Calendar.getInstance();

   @Override
   protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

      tvAlarmTime = findViewById(R.id.tvAlarmTime);
      Button btnSetTime = findViewById(R.id.btnSetTime);
      alarmManager = (AlarmManager) getSystemService(Context.ALARM_SERVICE);

       btnSetTime.setOnClickListener(v  -> {
          int hour = calendar.get(Calendar.HOUR_OF_DAY);
int minute = calendar.get(Calendar.MINUTE);

          TimePickerDialog timePickerDialog = new TimePickerDialog(MainActivi ty.this,
(TimePicker view, int selectedHour, int selectedMinute)  -> {
                    calendar.set(Calendar.HOUR_OF_DAY, selectedHour);
                    calendar.set(Calendar.MINUTE, selectedMinute);
                    calendar.set(Calendar.    SECOND, 0);

                    tvAlarmTime.setText("Alarm set for: " + selectedHour + ":"
+ String.format("%02d", selectedMinute));
                    setAlarm(calendar);
                }, hour, minute, true);
          timePickerDialog.show();
       });
   }

   private void setAlarm(Calendar targetCal) {

       Intent intent = new Intent(this, AlarmReceiver.class);
       pendingIntent = PendingIntent.getBroadcast(this, 0, intent,
PendingIntent.  FLAG_IMMUTABLE);
```

```
        alarmManager.set(AlarmManager.RTC_WAKEUP, targetCal.getTimeInMillis(),
pendingIntent);

        Toast.makeText(this, "Alarm Scheduled",
Toast.LENGTH_SHORT).show();     }
}
```

**AlarmReceiver.java**

```
package com.example.alarmapp;

//package com.example.alarmapp;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.Toast;

public class AlarmReceiver extends BroadcastReceiver {
    @Override
    public void onReceive(Context context, Intent intent) {
        Toast.makeText(context, "Alarm! Wake up!",
Toast.LENGTH_LONG).show();         // Can also trigger a notification here
    }
}
```

**AndroidManifest.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"

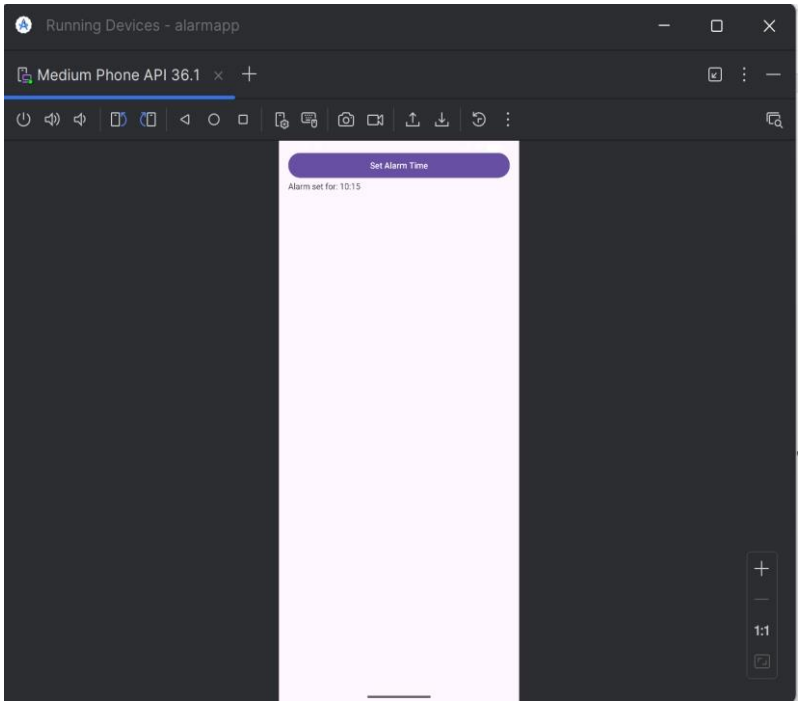
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Alarmapp">
        <activity
            android:name=".MainActivity"
            android:exported="true">
```

```
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />

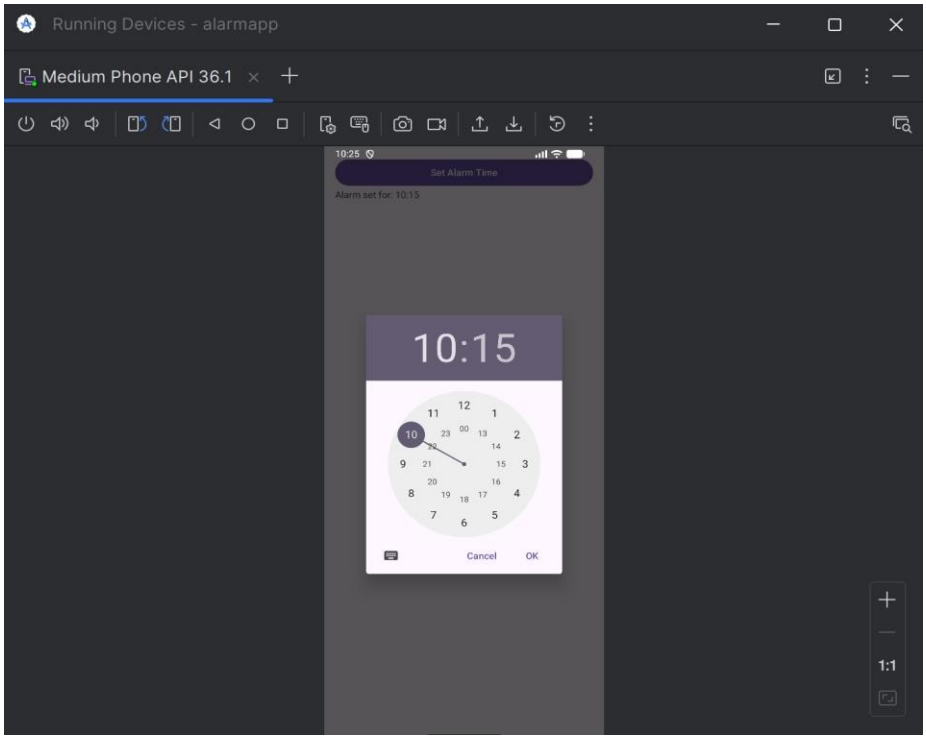
            <category android:name="android.intent.category.LAUNCHER"
/>          </intent -filter>
        </activity>
        <receiver android:name=".AlarmReceiver"/>
    </application>


</manifest>
```

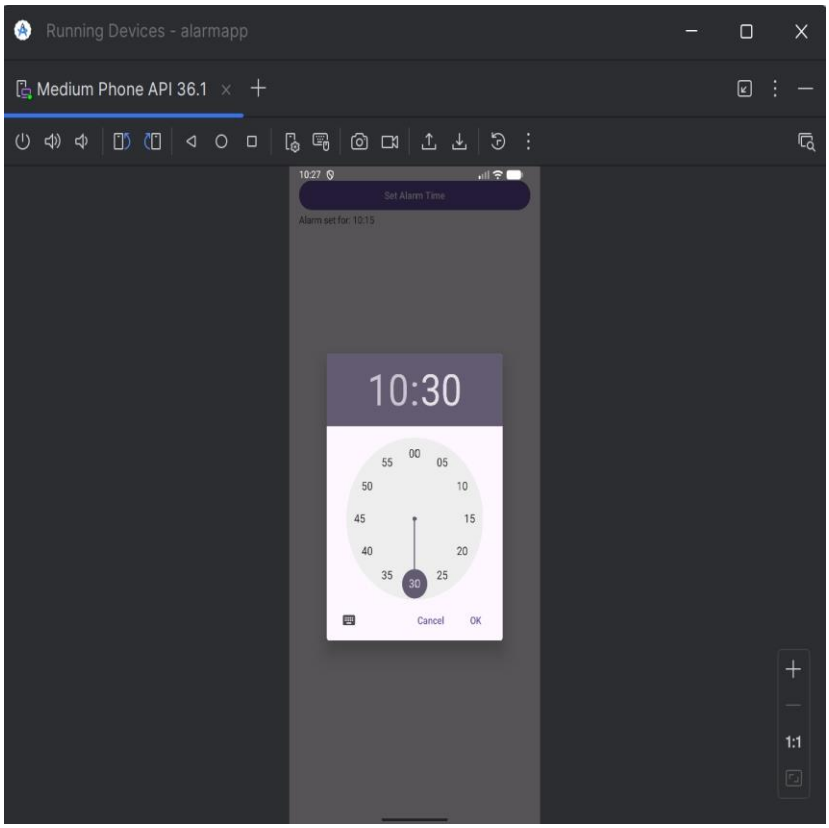
Output:

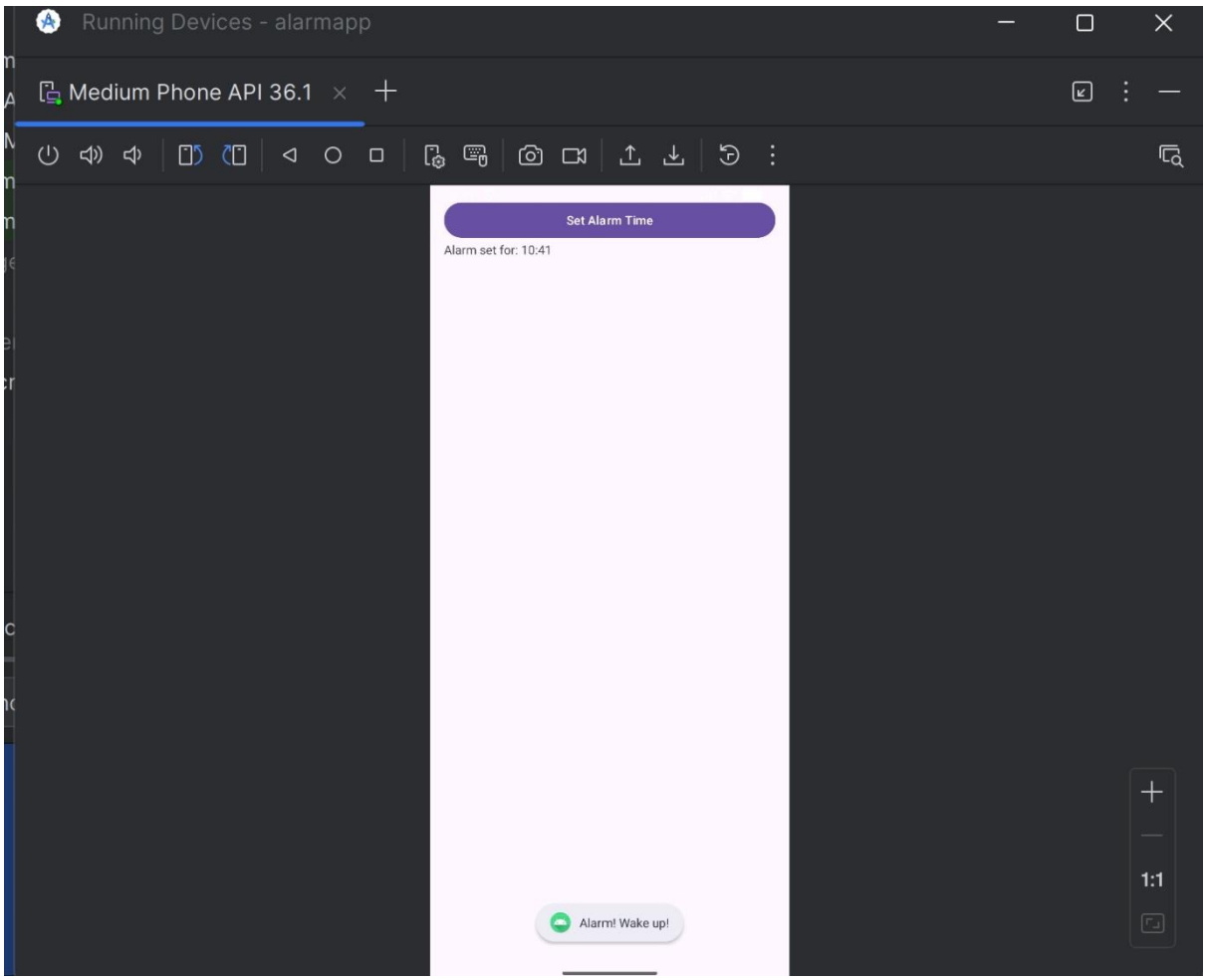


Hours:




Minitues:




Running Devices - alarmapp
Medium Phone API 36.1
Set Alarm Time
Alarm set for: 10:41
Alarm! Wake up!
1:1

Week-10

Create an application that shows the given URL (from a text field) in a browser

**AndroidManifest.xml**

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
package="com.example.openurlapp">

    <application
        android:allowBackup="true"
        android:label="Open URL App"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.NoActionBar">

        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category
android:name="android.intent.category.LAUNCHER"/>            </intent -
filter>
        </activity>

    </application>

</manifest>
```

**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation ="vertical"
   android:padding="24dp"
   android:gravity="center">

   <EditText
      android:id="@+id/editTextUrl"
      android:layout_width ="match_parent"
      android:layout_height ="wrap_content"
      android:hint="Enter URL (e.g. https://www.google.com)"
android:inputType="textUri"
      android:layout_marginBottom ="16dp" />

   <Button
      android:id="@+id/btnOpenUrl"
      android:layout_width ="wrap_content"
android:layout_height ="wrap_content"
android:text="Open in Browser"
      android:backgroundTint="#4CAF50"
      android:textColor="#FFFFFF"
      android:padding="10dp" />

</LinearLayout>
```

**MainActivity.java**

```
package com.example.openurlapp;


import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;

import android.net.Uri;

import android.os.Bundle;
```

```
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

   EditText     editTextUrl;
   Button btnOpenUrl;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

      editTextUrl = findViewById(R.id.editTextUrl);
      btnOpenUrl = findViewById(R.id.btnOpenUrl);

       btnOpenUrl.setOnClickListener(v   -> {
          String url = editTextUrl.getText().toString().trim();

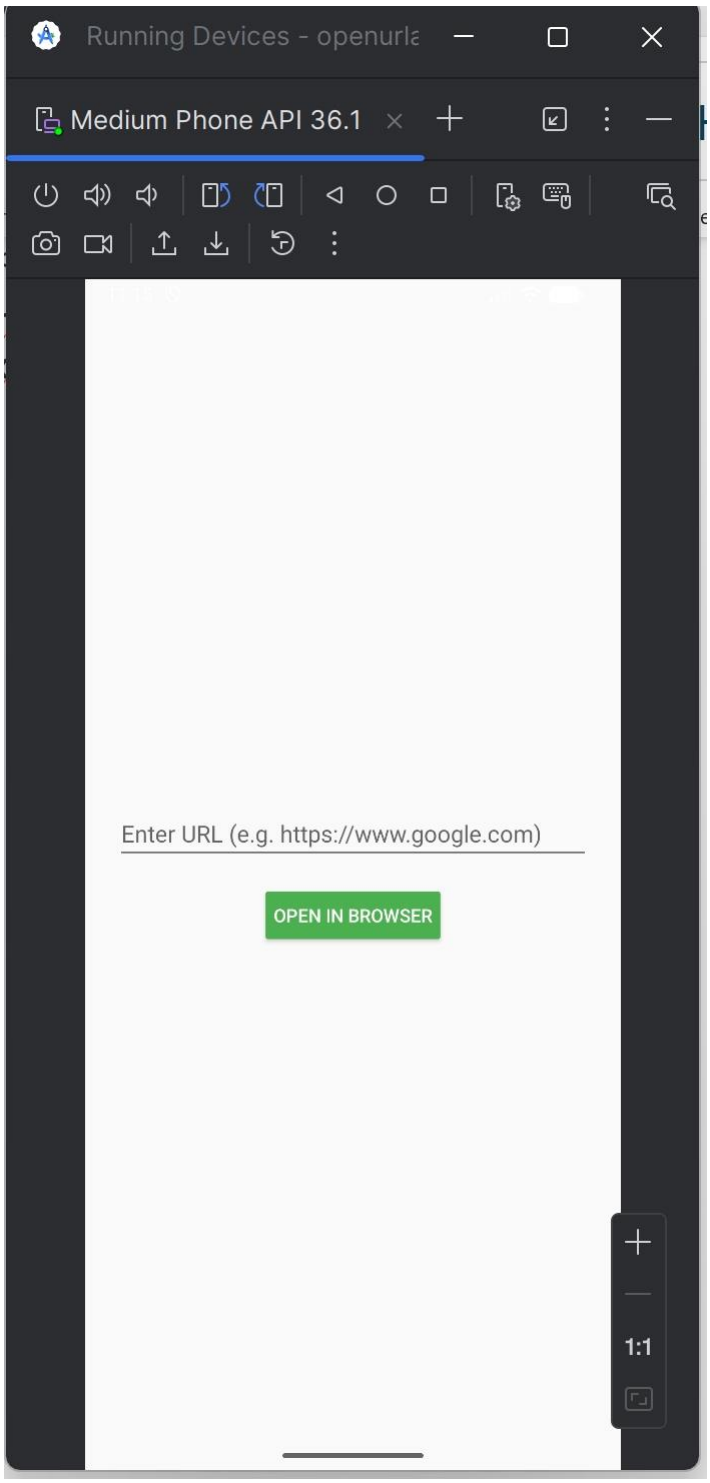
        if (url.isEmpty()) {
             Toast.makeText(MainActivity.this, "Please enter a
URL", Toast.LENGTH_SHORT).show();
             return;
}

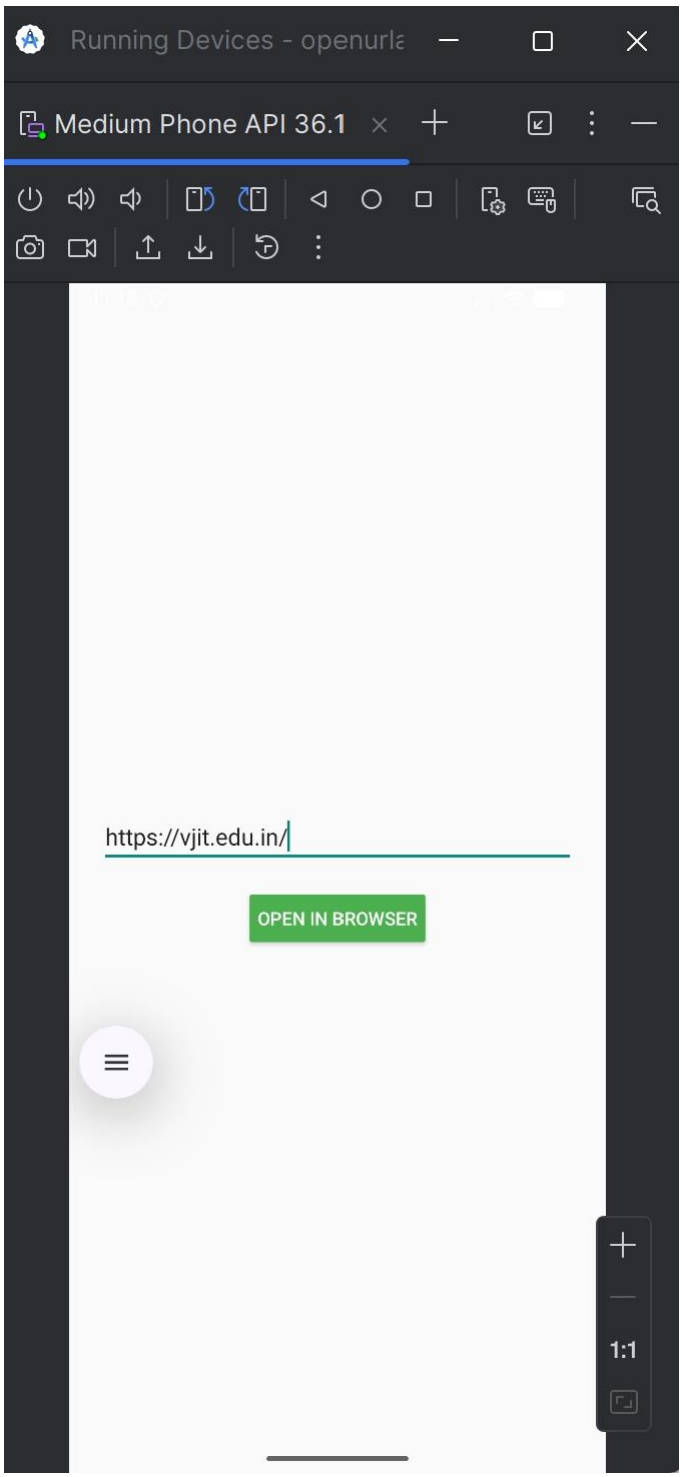
        // Add https:// if missing
        if (!url.startsWith("http://") && !url.startsWith("https://")) {
```

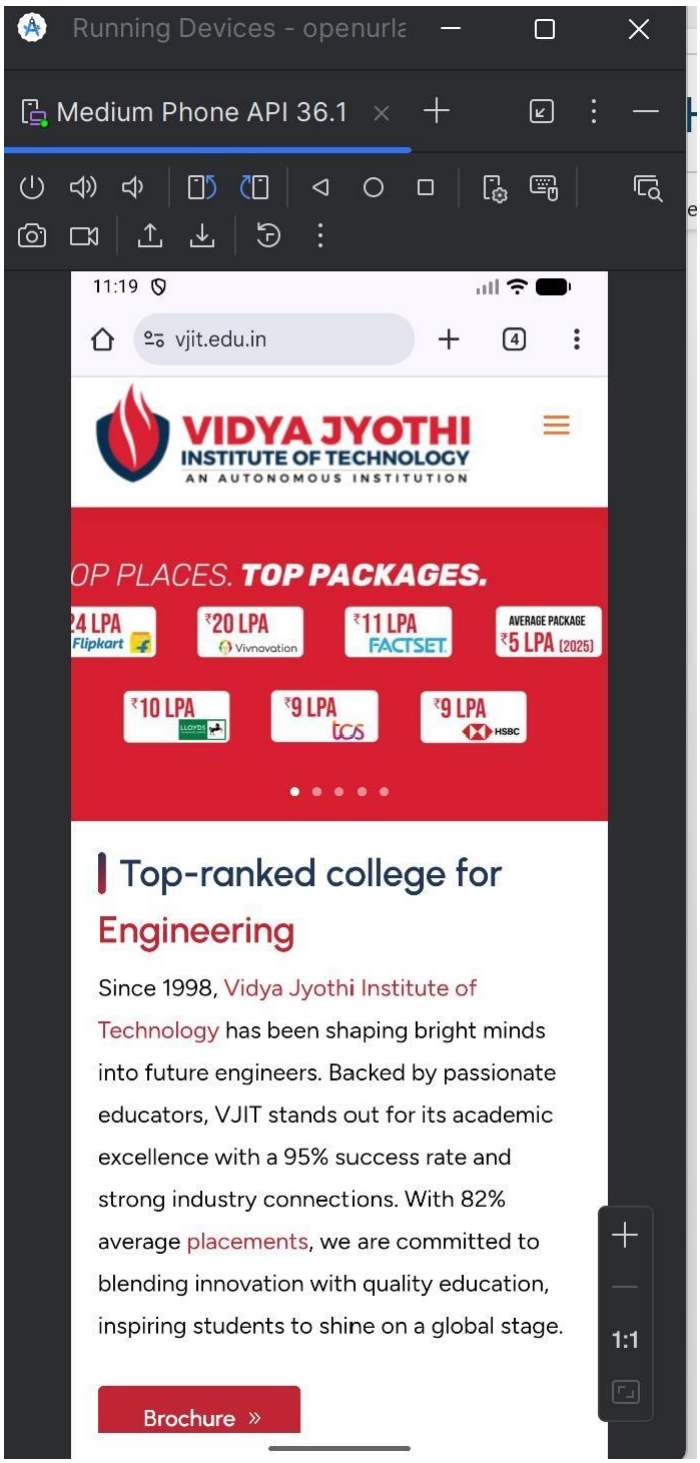
```
            url = "https://" + url;
}


        try {
            Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse(url));
startActivity(intent);
        } catch (Exception e) {
            Toast.makeText(MainActivity.this, "Invalid
URL!", Toast.LENGTH_SHORT).show();
        }
});    }
}
```

**Output:**


Running Devices - openurla
Medium Phone API 36.1
Enter URL (e.g. https://www.google.com)
OPEN IN BROWSER
1:1


Running Devices - openurla
Medium Phone API 36.1
https://vjit.edu.in/
OPEN IN BROWSER
1:1


Running Devices - openurla
Medium Phone API 36.1
11:19
vjit.edu.in
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Flipkart
₹20 LPA
Vivnovation
₹11 LPA
FACTSET
AVERAGE PACKAGE
₹5 LPA (2025)
₹10 LPA
₹9 LPA
tcs
₹9 LPA
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Since 1998, Vidya Jyothi Institute of Technology has been shaping bright minds into future engineers. Backed by passionate educators, VJIT stands out for its academic excellence with a 95% success rate and strong industry connections. With 82% average placements, we are committed to blending innovation with quality education, inspiring students to shine on a global stage.
Brochure »
1:1

Week-11

Develop a native application that uses GPS location information.

**activity_main.xml**

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp">

   <Button
      android:id="@+id/btnGetLocation"
      android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Get Location" />

   <TextView
      android :id="@+id/tvLocation"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Location will appear here"
android:textSize="18sp"/> </LinearLayout>
```

**MainActivity.java**

```
package com.example.gpslocationapp;

//package com.example.gpslocationapp;

import android.Manifest;
import android.annotation.SuppressLint;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationListener;
import android.location.LocationManager;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
```

```
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {
   private static final int  LOCATION_PERMISSION_CODE = 100;
private LocationManager locationManager;
   private TextView tvLocation ;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

      Button btnGetLocation = findViewById(R.id.btnGetLocation);
      tvLocation = findViewById(R.id.tvLocation );

       btnGetLocation.setOnClickListener(v -> getLocation());
}

   private void getLocation () {
      if (ContextCompat.checkSelfPermission(this,
Manifest.permission.ACCESS_FINE_LOCATION) !=
PackageManager. PERMISSION_GRANTED) {
          ActivityCompat.requestPermissions(this, new
String[]{Manifest.permission.  ACCESS_FINE_LOCATION},
LOCATION_PERMISSION_CODE);
      } else {
          locationManager = (LocationManager) getSystemService(LOCATION_SERVICE);
locationManager.requestLocationUpdates(LocationManager.GPS_PROVIDER, 0, 0,
locationListener);
      }
   }

   private final LocationListener locationListener = new LocationListener() {
@SuppressLint ("SetTextI18n")
      @Override
      public void onLocationChanged(Location location) {
          double  latitude = location.getLatitude();
          double  longitude = location.getLongitude();

          tvLocation.setText("Latitude: " + latitude + "\nLongitude: " + longitude);
          locationManager.removeUpdates(this); // Optional: remove after first update
```

```
        }
        @Override public void onStatusChanged(String provider, int status, Bundle extras) {}
@Override public void onProviderEnabled (@NonNull String provider) {}
        @Override public void onProviderDisabled(@NonNull String provider) {
            Toast.makeText(MainActivity.this, "Please enable
GPS", Toast.LENGTH_LONG).show();
        }
    };

    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[]
permissions, @NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
if (requestCode ==  LOCATION_PERMISSION_CODE) {
            if (grantResults.  length > 0 && grantResults[0] ==
PackageManager. PERMISSION_GRANTED) {
                getLocation();
            } else {
                Toast.makeText(this, "Location permission
denied", Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```

**AndroidManifest.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"
/>
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"

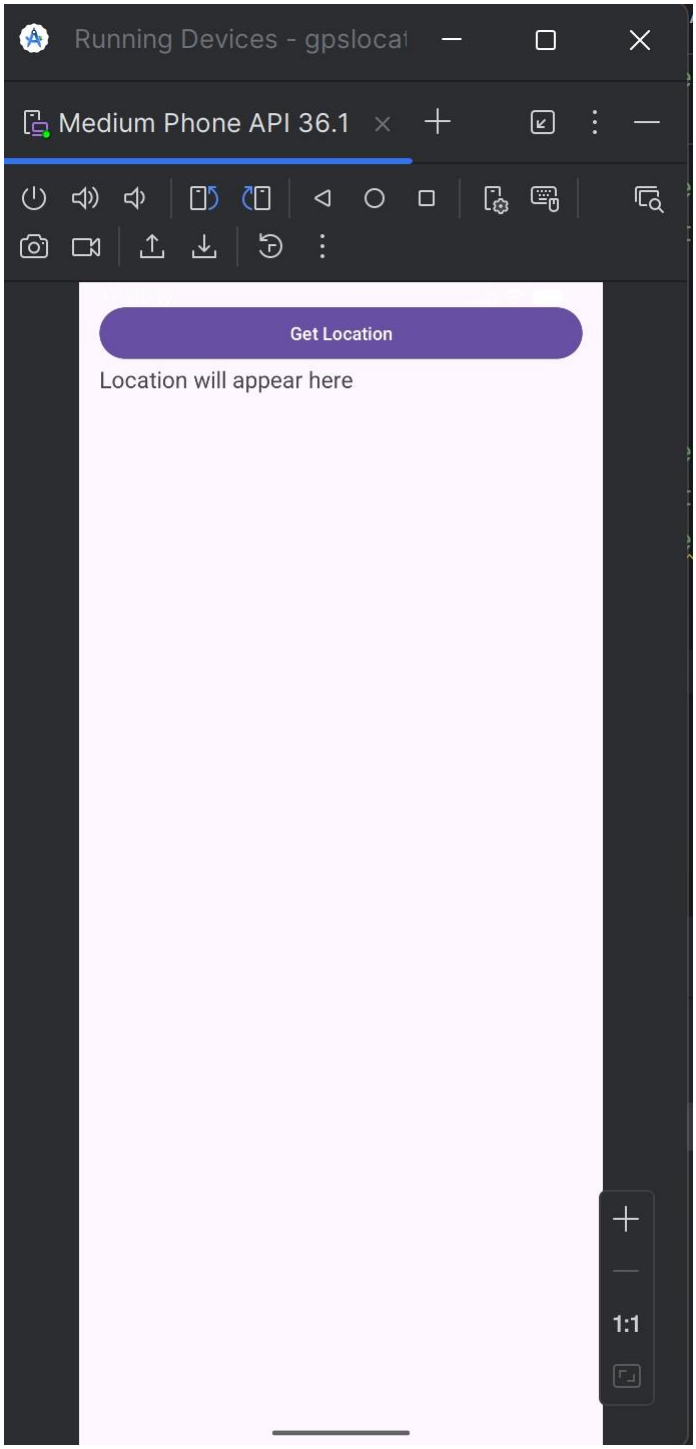
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Gpslocationapp">
        <activity
```

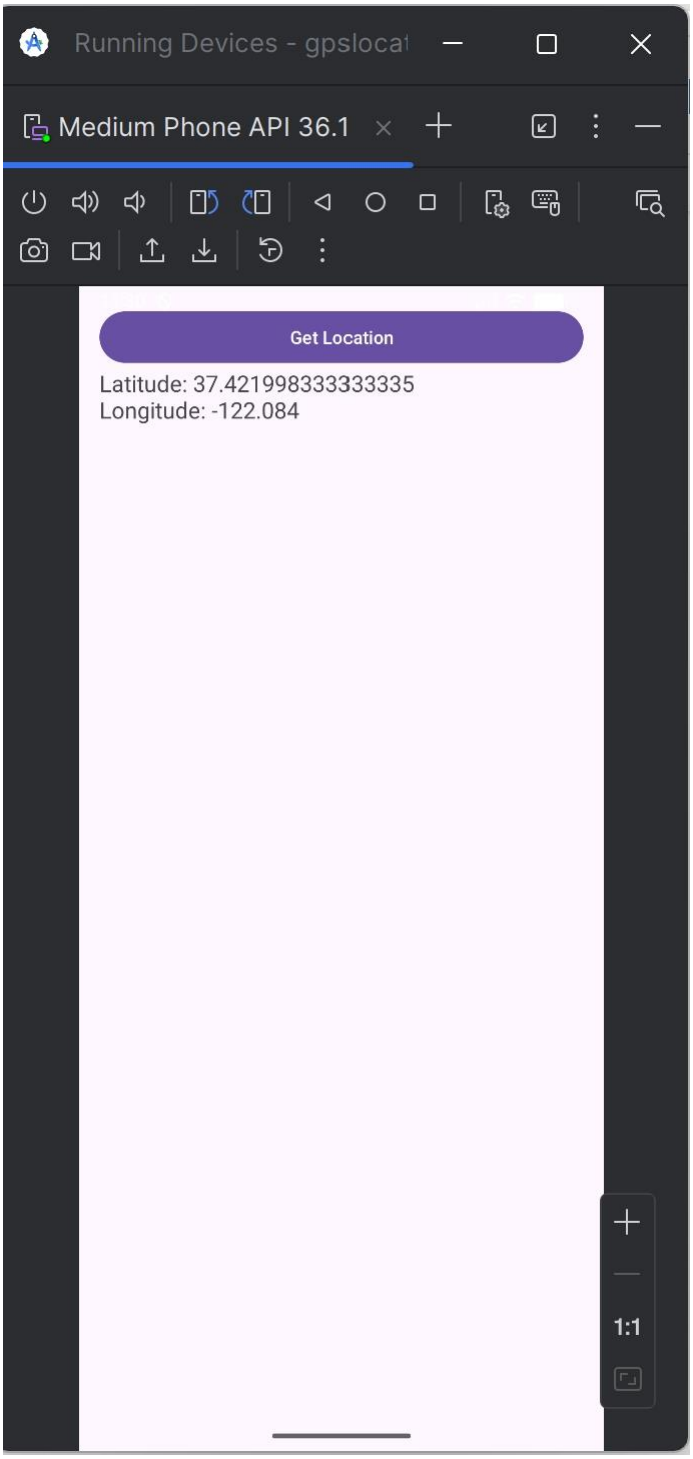
```
        android :name =".MainActivity"
        android :exported ="true">
        <intent -filter>
            <action android:name="android.intent.action.MAIN" />

            <category android:name="android.intent.category.LAUNCHER" />
</intent -filter>
      </activity>
   </application>

</manifest>
```

Outputs:


Running Devices - gpsloca
Medium Phone API 36.1
Get Location
Location will appear here
1:1


Running Devices - gpsloca
Medium Phone API 36.1
Get Location
Latitude: 37.421998333333335
Longitude: -122.084
1:1